AI-BASED SELF-EVALUATION SYSTEM FOR COMMUNICATION SKILLS USING OPENCV, NLP, AND SPEECH VALIDATION

A MINI PROJECT REPORT

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ABSTRACT

Effective interpersonal communication represents an essential determinant of professional, academic, and career success. Whether participating in competitive placement interviews, delivering technical presentations, or engaging in professional discourse, candidates are rigorously evaluated on both verbal and non-verbal behaviors. Traditional communication coaching and mock interview assessments rely predominantly on manual evaluations conducted by human trainers. However, human evaluation is inherently constrained by subjective grading inconsistencies, observer fatigue, high logistical scheduling friction, and an inability to provide synchronous, multi-dimensional tracking during self-paced practice.

To overcome these foundational limitations, this project presents the design, development, and experimental validation of an **AI-Based Self-Evaluation System for Communication Skills Using OpenCV, NLP, and Speech Validation**. The developed framework delivers an integrated, real-time multimodal evaluation platform that synchronizes computer vision, speech acoustics, and natural language processing into a unified feedback architecture. Using OpenCV alongside MediaPipe Face Mesh, Pose, and Hands pipelines, the system tracks 468 dense facial landmarks and skeletal joints at 30 frames per second to evaluate critical non-verbal behavioral cues, including eye contact percentage via horizontal iris-to-screen center offset, head pose stability through nose-to-face midline yaw estimation, spinal posture alignment via bilateral shoulder elevation deviation, smiling dynamics with temporal moving average filtering, and hand gesture frequency.

Simultaneously, a multi-threaded audio acquisition engine captures raw speech at 16 kHz mono, applies Butterworth bandpass filtering, and performs automated speech transcription using the OpenAI Whisper transformer architecture. The extracted word timestamps and transcript undergo comprehensive linguistic and acoustic evaluation to quantify speaking rate (Words Per Minute), pause and silence duration, hesitation and filler-word density (such as *um*, *uh*, *like*, and *actually*), grammatical syntax coherence, and lexical diversity. An automated multimodal scoring engine fuses visual and speech metrics into an interpretable 0–100 overall communication score, categorized into objective proficiency ratings accompanied by transparent strengths, targeted improvement areas, and actionable pedagogical suggestions. Experimental validation on multiple test subjects demonstrates strong tracking fidelity, high transcription accuracy (WER < 6.8%), robust real-time throughput, and effective performance improvement across repeated practice sessions.

Keywords: Communication Skills, OpenCV, MediaPipe, Computer Vision, Speech Recognition, Whisper STT, Natural Language Processing, Multimodal Evaluation, Acoustic Analysis, Self-Paced Learning.

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ABBREVIATION

Abbreviation	Expansion
AI	Artificial Intelligence
API	Application Programming Interface
ASR	Automatic Speech Recognition
B.Tech	Bachelor of Technology
CPU	Central Processing Unit
CSV	Comma-Separated Values
CV	Computer Vision
DFD	Data Flow Diagram
EWPM	Effective Words Per Minute
FPS	Frames Per Second
GPU	Graphics Processing Unit
GUI	Graphical User Interface
HOD	Head of the Department
IEEE	Institute of Electrical and Electronics Engineers
JSON	JavaScript Object Notation
ML	Machine Learning
NLP	Natural Language Processing
OpenCV	Open Source Computer Vision Library
RGB	Red, Green, Blue
STT	Speech-to-Text
UI	User Interface
VAD	Voice Activity Detection
WAV	Waveform Audio File Format
WER	Word Error Rate
WPM	Words Per Minute

CHAPTER 1

INTRODUCTION

Effective communication serves as the fundamental cornerstone of interpersonal interaction, professional advancement, academic excellence, and leadership across contemporary engineering and organizational domains. In professional spheres such as corporate interviews, technical presentations, managerial evaluations, and public discourse, success depends not merely upon subject matter expertise, but critically upon the speaker's ability to articulate concepts with clarity, composure, and confidence. Communication represents an intricate, multimodal process comprising both verbal dynamics—such as grammatical coherence, vocabulary richness, discourse organization, and semantic relevance—and non-verbal cues—such as sustained eye contact, stable head posture, erect body alignment, responsive facial expressions, and natural hand gesture dynamics.

Despite the recognized criticality of effective communication, students, graduates, and aspiring professionals frequently encounter severe challenges in mastering these competencies. Developing exemplary communication skills demands repeated practice, continuous self-reflection, and objective, constructive feedback. However, conventional training methodologies rely almost exclusively on human evaluations conducted by specialized communication trainers, interview panelists, or faculty mentors. While human assessments provide intuitive contextual insights, they present deep structural bottlenecks: they are inherently subjective, prone to observer fatigue, logistically expensive, and incapable of providing continuous, synchronized, quantitative feedback during self-paced solitary practice.

Recent breakthroughs in artificial intelligence, computer vision, speech acoustics, and natural language processing (NLP) offer an unprecedented opportunity to democratize, automate, and objectify communication skills assessment. State-of-the-art computer vision libraries, notably OpenCV and MediaPipe, enable high-precision, real-time spatial tracking of facial landmarks, iris orientation, spinal posture, and upper-limb gestures directly from commodity webcams without dedicated sensor hardware. Concurrently, advancements in deep neural automatic speech recognition (ASR)—exemplified by OpenAI's Whisper transformer architecture—allow robust, timestamped speech transcription even in the presence of conversational disfluencies and ambient acoustic noise.

This project introduces an end-to-end **AI-Based Self-Evaluation System for Communication Skills Using OpenCV, NLP, and Speech Validation**. The developed framework delivers an integrated, real-time multimodal evaluation platform that synchronizes computer vision, speech acoustics, and natural language processing into a unified feedback architecture. By simultaneously monitoring visual behavior at 30 frames per second and capturing acoustic signals at 16 kHz mono, the system computes objective metrics across eye contact, head orientation stability, skeletal posture alignment, facial expression dynamics, hand gesture engagement, speaking rate (WPM), acoustic pause duration, filler word frequency, and grammatical correctness.

1.1 COMMUNICATION DYNAMICS IN ACADEMIC AND PROFESSIONAL DOMAINS

Communication is fundamentally a multi-channel transmission of information, intent, and emotion. In high-stakes professional evaluations, such as campus placement interviews and technical defense presentations, evaluators synthesize verbal signals with non-verbal physical manifestations. Verbal dimensions encompass grammatical correctness, vocabulary richness, structural coherence, and topical relevance. Non-verbal dimensions encompass gaze orientation, facial expressiveness, head motion stability, upper-body posture, and hand gesture kinematics.

Research in behavioral psychology and pedagogical communication demonstrates that non-verbal cues account for a dominant proportion of the listener's perception of speaker confidence and credibility. For instance, maintaining regular eye contact with an audience or camera fosters trust and projects authority. Conversely, averted gaze, erratic head tilt, or slumped posture signals anxiety, cognitive overburden, or lack of preparation. Similarly, in speech

acoustics, an optimal speaking rate between 120 and 160 words per minute (WPM) conveys fluency, whereas excessive pauses (> 3.0 seconds) and high filler-word frequencies (e.g., *um*, *uh*, *like*, *basically*) significantly degrade listener engagement.

Consequently, an effective communication evaluation tool must operate as a multimodal observer. Evaluating speech transcripts in isolation ignores body language and eye contact; conversely, monitoring video streams without speech analysis ignores linguistic fluency and factual coherence. The integration of computer vision, speech recognition, and natural language processing resolves this dichotomy, establishing a holistic, quantitative evaluation paradigm.

1.2 EVOLUTION OF COMMUNICATION ASSESSMENT & MOCK INTERVIEW SYSTEMS

Communication assessment has evolved across three major paradigms over recent decades:

1. Manual Human Evaluation: Historically, public speaking clubs, campus placement cells, and corporate training programs have relied on face-to-face mock interviews and group discussions evaluated by faculty or industry practitioners. While human evaluators offer qualitative empathy, the evaluation is constrained by subjective bias, differing evaluator standards, observational fatigue, and high per-session scheduling costs. Furthermore, manual assessment lacks granular quantitative metrics; an evaluator cannot accurately count the exact number of seconds a student maintained eye contact or measure the precise millisecond duration of hesitation pauses.

2. Monomodal Software Systems: With the advent of personal computers and digital audio-video recording, early software platforms introduced video playback and basic acoustic recording. Subsequent systems introduced automated speech-to-text or grammar checking in isolated text editors. However, these systems remained uncoupled; learners received disjointed reports on text grammar without any non-verbal visual assessment, or received raw video recordings without automated acoustic or linguistic scoring.

3. Intelligent Multimodal AI Platforms: Modern intelligent evaluation systems utilize deep learning, real-time landmark tracking, and transformer-based speech engines to observe, quantify, and fuse verbal and non-verbal communication behaviors synchronously. The system developed in this project represents this cutting edge, processing live video frames and microphone audio streams locally to generate immediate, multi-dimensional feedback without requiring cloud data transmission or manual intervention.

1.3 ROLE OF COMPUTER VISION, SPEECH PROCESSING, AND NLP

The convergence of computer vision, speech processing, and natural language processing enables automated communication assessment to achieve high fidelity and reliability:

Computer Vision: OpenCV provides the optimized framework for video capture, frame manipulation, color space transformation, and graphical rendering. MediaPipe provides lightweight, machine learning-driven perception pipelines that execute at real-time speeds on standard CPUs. Specifically, MediaPipe Face Mesh extracts 468 dense 3D facial landmarks, enabling sub-pixel tracking of iris coordinates (landmarks 468 and 473) for gaze and eye contact estimation, as well as lip contour coordinates for smiling detection. MediaPipe Pose tracks 33 skeletal landmarks to quantify bilateral shoulder tilt for posture evaluation, while MediaPipe Hands monitors 21 keypoints per hand to quantify purposeful gesture usage.

Speech Processing and Automatic Speech Recognition (ASR): Capturing acoustic signals at 16 kHz mono through multi-threaded audio buffers ensures crisp audio recording. Signal preprocessing applies Butterworth bandpass filtering (80 Hz to 7500 Hz) to eliminate low-frequency environmental rumble and high-frequency electrical hiss. The OpenAI Whisper speech-to-text transformer model transcribes speech into text while yielding precise word-level start and end timestamps. These timestamps allow direct mathematical calculation of speaking rate (WPM), speech-to-silence ratios, and millisecond-accurate pause durations.

Natural Language Processing: NLP components analyze the transcribed text for grammatical syntax coherence, clause structure, lexical redundancy, and filler word density. By scanning transcripts against conversational disfluency lexicons and grammatical patterns, NLP provides transparent, rule-based feedback on linguistic delivery.

1.4 PROBLEM STATEMENT

In contemporary engineering education and professional training, students and job seekers face significant hurdles in objectively preparing for interviews, technical presentations, and communication assessments. Traditional manual evaluation mechanisms suffer from four fundamental shortcomings:

1. Subjectivity and Inconsistent Scoring: Human evaluators possess varied standards, expectations, and personal biases. A candidate may receive contradictory ratings from different interviewers for identical communication delivery, preventing structured, measurable progress tracking.

2. High Logistical Cost and Scheduling Constraints: Organizing one-on-one mock interviews with experienced human trainers requires significant faculty time, physical infrastructure, and scheduling coordination. Consequently, students receive at most one or two practice sessions prior to campus recruitment drives, which is grossly inadequate for meaningful behavioral improvement.

3. Lack of Real-Time, Multi-Dimensional Tracking: Human observers experience cognitive overload when tasked with simultaneously monitoring eye contact, spinal posture, facial micro-expressions, speech pacing, acoustic pauses, filler words, and grammatical correctness. Important delivery flaws are routinely overlooked or vaguely reported without quantitative evidence.

4. Disconnected Verbal and Non-Verbal Feedback: Existing automated practice applications typically analyze either text alone (e.g., automated grammar checkers) or video alone (e.g., basic face detection tools), leaving a critical void for an integrated, real-time multimodal evaluation platform capable of fusing verbal acoustics and non-verbal physical behavior into a single, cohesive feedback framework.

Therefore, the core problem addressed by this project is:

"To design, implement, and validate an intelligent, real-time multimodal self-evaluation system that combines OpenCV computer vision, MediaPipe landmark tracking, Whisper speech transcription, and NLP acoustic analysis to deliver objective, synchronous, and actionable communication skill assessment."

1.5 OBJECTIVES OF THE STUDY

The primary objective of this project is to develop a standalone, multi-threaded software platform that continuously evaluates, scores, and coaches users on their verbal and non-verbal communication performance in real time.

The specific technical objectives are:

1. To design and deploy a real-time computer vision pipeline utilizing OpenCV and MediaPipe Face Mesh to track 468 facial landmarks and derive quantitative metrics for eye contact and head pose stability.
2. To incorporate MediaPipe Pose and Hands pipelines to continuously monitor spinal posture alignment via bilateral shoulder elevation and detect upper-limb hand gesture activity.
3. To implement a facial expression analysis module utilizing normalized mouth width and lip corner lift ratios with temporal moving average filtering for smooth smile detection.
4. To develop a multi-threaded acoustic capture pipeline capable of recording 16 kHz audio, applying digital bandpass filtering, and transcribing speech into timestamped text via OpenAI Whisper.

5. To build an NLP and acoustic analytics engine that computes speaking rate (Words Per Minute), detects hesitation pauses (> 1.5s), identifies conversational filler words, and evaluates grammatical coherence.
6. To design a weighted multimodal scoring formulation that aggregates vision metrics (0–100 scale) and speech metrics (0–10 scale) into standardized proficiency tiers (Excellent, Good, Average, Needs Improvement).
7. To develop an automated feedback generator that produces actionable pedagogical strengths, targeted improvement areas, and practical coaching recommendations.
8. To construct an intuitive, dark-mode Graphical User Interface using CustomTkinter with embedded Matplotlib analytical trend charts and local JSON/TXT report persistence.

1.6 SCOPE OF THE PROJECT

The scope of this project encompasses the design, implementation, and experimental evaluation of an AI-based communication self-evaluation platform:

• The system operates using standard consumer-grade webcams and built-in or external microphones, eliminating the need for expensive motion capture suits, multi-camera arrays, or specialized biometric hardware.
• Video analysis processes RGB frames at 30 frames per second at 640x480 resolution, balancing high tracking precision with low CPU utilization.
• Audio processing captures 16-bit mono PCM audio at 16,000 Hz, applying Butterworth filtering to isolate fundamental human vocal frequencies.
• The software architecture is engineered entirely in Python, integrating OpenCV, MediaPipe, NumPy, SciPy, SoundDevice, Whisper, and CustomTkinter.
• Complete data privacy is preserved: all video frame processing, landmark tracking, audio transcription, and score computation execute locally on the candidate's machine without transmitting video or voice data to third-party cloud servers.
• The project is evaluated across multiple candidate sessions, comparing metrics before and after practice, and generating exportable evaluation reports.

1.7 METHODOLOGY OVERVIEW

The project follows a structured seven-phase closed-loop engineering methodology:

Phase 1: Hardware & Environment Initialization: The candidate launches the CustomTkinter GUI. The system initializes multi-threaded camera acquisition (OpenCV VideoCapture) and background Whisper speech model loading.
Phase 2: Video Stream Landmark Extraction: Captured video frames are converted to RGB and piped through MediaPipe Face Mesh, Pose, and Hands. Sub-pixel 3D landmarks are extracted for irises (468, 473), nose tip (1), face boundaries (234, 454), lips (61, 291, 0, 17), shoulders (11, 12), and wrists/fingertips.
Phase 3: Real-Time Non-Verbal Metric Computation: Distance formulas compute eye contact percentage; horizontal offsets classify head yaw (Straight, Left, Right); bilateral shoulder elevation differences quantify posture (Good, Average, Poor); lip geometry ratios compute smile scores; hand presence computes gesture scores.
Phase 4: Threaded Audio Recording & Whisper Transcription: When the candidate initiates a practice session, SoundDevice records raw audio into a memory buffer. At session completion, digital bandpass filtering is applied, and the Whisper model produces a timestamped text transcript.
Phase 5: Speech & NLP Analytics: The speech analyzer parses word timestamps to calculate Words Per Minute (WPM), speech-to-pause ratios, long hesitation silences, filler word density, and syntactic grammar patterns.
Phase 6: Multimodal Score Fusion & Feedback Generation: The scoring engine computes weighted visual performance (0–100) and speech evaluation points (0–10). A rule-based feedback generator maps performance indicators to personalized strengths and suggestions.
Phase 7: Dashboard Visualization & Report Persistence: The UI updates real-time scorecards, renders Matplotlib historical performance trends, writes session records to `session_history.json`, and outputs formatted text reports.

1.8 SIGNIFICANCE AND IMPACT OF THE PROJECT

The proposed system provides transformative benefits across academic and professional sectors:

• **Anxiety-Free Independent Practice:** Candidates can practice interviews and presentations privately without social embarrassment or performance anxiety, building confidence iteratively.
• **Scalable Campus Placement Training:** Academic institutions can deploy the platform across hundreds of students simultaneously, overcoming trainer shortages and democratizing access to quality soft-skill coaching.
• **Objective, Repeatable Benchmarking:** Quantitative metrics eliminate subjective evaluator bias, providing transparent, standardized metrics that track true longitudinal skill improvement.
• **Zero-Cost Open-Source Stack:** Built entirely on open-source Python libraries, the solution incurs ₹0 licensing fees and requires no recurring subscription costs.

CHAPTER 2

LITERATURE REVIEW

The literature review provides a comprehensive analysis of prior research, technical frameworks, algorithmic methodologies, and empirical studies in intelligent communication skills assessment, affective computing, computer vision, and speech processing. Over the past two decades, automated behavioral assessment has transitioned from simple heuristic-based acoustic feature extraction to sophisticated multimodal deep learning architectures capable of synchronous visual, acoustic, and linguistic analysis.

2.1 INTRODUCTION

Human communication is inherently multimodal. In face-to-face interactions, listeners synthesize acoustic pitch and intonation, linguistic syntax and semantics, and non-verbal physical signals. Early computing attempts to evaluate communication focused on isolated modalities, such as automated essay scoring or basic speech recognition. However, modern research recognizes that effective communication coaching requires holistic observation. This chapter reviews foundational and contemporary advancements across three core areas: (1) computer vision techniques for non-verbal behavioral tracking, (2) automatic speech recognition and acoustic analysis, and (3) multimodal fusion frameworks for holistic evaluation.

2.2 EVOLUTION OF COMMUNICATION ASSESSMENT TECHNIQUES

Traditional communication evaluation relies on established human observation protocols, such as public speaking rating scales, Toastmasters evaluation guidelines, and standardized rubrics developed by the Association of American Colleges and Universities (AAC&U). While these rubrics identify essential behavioral constructs—such as delivery, posture, eye contact, and language—their execution relies on subjective human judgment.

With the rise of interactive computing, early automated systems explored virtual agents and computerized mock interviews. Systems like Mach (My Automated Conversation CoacH) developed at MIT demonstrated that users interacting with a 3D animated agent and receiving automated feedback on speech rate and facial smiles showed measurable improvements in interview performance. Subsequent systems incorporated machine learning classifiers to predict interview ratings based on multimodal feature vectors.

2.3 COMPUTER VISION FOR NON-VERBAL BEHAVIORAL ANALYSIS

Computer vision techniques enable non-invasive, objective measurement of visible physical behaviors. Non-verbal signals communicate confidence, composure, engagement, and attentiveness.

2.3.1 Facial Landmark Extraction & Morphology

Facial analysis has progressed from early Viola-Jones Haar cascade feature extractors to Active Appearance Models (AAM), Constrained Local Models (CLM), and deep convolutional landmark detectors. Bradski [1] established OpenCV as the universal framework for computer vision operations, providing optimized primitives for matrix transformation and frame processing. Lugaresi et al. [2] introduced MediaPipe, a perception pipeline architecture developed by Google. MediaPipe Face Mesh utilizes a lightweight deep neural network to infer 468 3D facial landmarks from a single RGB camera frame in real time. Unlike bounding-box face detectors, dense 3D landmarking allows exact geometric calculation of facial contours, eyelid openings, and lip corner displacement.

2.3.2 Gaze Tracking & Iris Proximity Quantification

Eye contact is universally recognized as a primary indicator of speaker confidence, credibility, and engagement. Traditional gaze estimation required specialized infrared eye trackers or complex calibration routines. Recent developments in deep landmark tracking incorporate refined iris landmarking (MediaPipe Iris), tracking 5 dedicated keypoints per eye, including the precise iris center (landmarks 468 and 473). By calculating the Euclidean distance between the detected iris center and the optical center of the video frame, systems can reliably estimate whether the speaker is maintaining gaze with the camera/audience or looking away.

2.3.3 Pose Estimation & Skeletal Posture Dynamics

Upright, balanced posture reflects poise, openness, and physical composure. In public speaking and interview settings, asymmetrical shoulder slouching or forward head tilt indicates tension, fatigue, or lack of confidence. Pose estimation frameworks such as OpenPose, PoseNet, and MediaPipe Pose track 33 3D full-body skeletal landmarks. By tracking landmark 11 (left shoulder) and landmark 12 (right shoulder), systems can continuously measure vertical elevation differences ($|y1 - y2|$), classifying posture into distinct stability categories in real time.

2.3.4 Hand Gesture Detection in Public Speaking

Purposeful hand gestures reinforce spoken ideas, enhance vocal emphasis, and clarify spatial concepts. Conversely, nervous fidgeting, hiding hands, or excessive, chaotic waving can distract listeners. MediaPipe Hands provides high-fidelity tracking of 21 3D landmarks per hand. Tracking the presence and kinematic activity of one or both hands allows automated systems to quantify gesture frequency and encourage natural, controlled gesturing.

2.4 AUTOMATIC SPEECH RECOGNITION & ACOUSTIC PROCESSING

Verbal delivery is evaluated through both what is said (linguistic content) and how it is said (acoustic delivery).

2.4.1 Speech-to-Text: From HMMs to Whisper Transformers

Early Automatic Speech Recognition (ASR) relied on Hidden Markov Models (HMMs) combined with Gaussian Mixture Models (GMMs). While computationally efficient, these models struggled with varied accents, background noise, and conversational disfluencies. Recent end-to-end deep learning architectures, such as Baevski et al.'s wav2vec 2.0 [5] and OpenAI's Whisper, revolutionized speech recognition. Trained on over 680,000 hours of multilingual, diverse web audio, Whisper demonstrates remarkable robustness to ambient noise, microphone variability, and regional accents. Furthermore, Whisper outputs word-level timestamps, enabling precise correlation between acoustic timing and textual transcripts.

2.4.2 Speaking Rate (WPM) & Pause Quantification

Speaking rate, measured in Words Per Minute (WPM), directly impacts intelligibility and listener comprehension. Speech communication research indicates that an ideal conversational and presentation delivery rate ranges between 120 and 160 WPM. Delivery exceeding 190 WPM sounds rushed and anxious, whereas delivery below 100 WPM appears sluggish or hesitant. Furthermore, acoustic pauses provide vital pacing; however, prolonged silent pauses exceeding 2.5 to 3.0 seconds often reflect cognitive freezing or lack of preparation.

2.4.3 Disfluency Modeling & Filler Word Detection

Conversational disfluencies—such as filler words (*um*, *uh*, *er*, *like*, *you know*, *basically*, *actually*) and adjacent word repetitions (stuttering)—frequently manifest when speakers experience cognitive load or verbal anxiety. Excessive filler word usage diminishes perceived authority and distracts audiences. Automated NLP parsing maps transcripts against curated conversational disfluency lexicons to quantify filler ratios and highlight repetitive verbal crutches.

2.5 NATURAL LANGUAGE PROCESSING FOR VERBAL PERFORMANCE

Jurafsky and Martin [3] outline the foundational principles of natural language processing for computational discourse analysis. In communication coaching, NLP methods evaluate:

• **Grammatical Syntax:** Identifying subject-verb agreement slips, tense mismatches, and fragmented clauses.
• **Lexical Diversity:** Measuring the Type-Token Ratio (TTR) and vocabulary sophistication to prevent monotonous repetition.
• **Sentence Structure:** Encouraging balanced sentence lengths and structured discourse with clear introductory, explanatory, and concluding elements.

2.6 MULTIMODAL INFORMATION FUSION IN BEHAVIORAL COMPUTING

Multimodal fusion strategies are categorized into early fusion (feature-level), late fusion (decision-level), and hybrid fusion. In behavioral assessment, late fusion is widely recognized as superior for explainability and pedagogical feedback. By calculating independent, validated scores for vision (eye contact, posture, gestures) and speech (fluency, grammar, clarity), the system can present transparent, inspectable component scores rather than an opaque black-box aggregate.

2.7 EXISTING SYSTEMS & COMPARATIVE RESEARCH CONTRIBUTIONS

Recent academic and commercial initiatives reflect the growing interest in automated interview coaching:

• *AI-Enabled Automated Interview Evaluation System* (IJIRMPS, 2025): Integrated NLP semantic scoring with basic OpenCV face verification, but lacked continuous gaze, posture, and acoustic pause tracking.
• *InterVista: AI Virtual Interview System* (Sharma et al., IJPREMS, 2025): Combined NLP question generation with facial expression recognition, but relied on cloud API inference, creating latency and privacy concerns.
• *AI-Enhanced Confident Speaker* (IJERT, 2026): Explored multimodal ML models for public speaking, demonstrating the effectiveness of combining acoustic and visual indicators.
• *Commercial Platforms (e.g., Yoodli, Orai, Speeko):* Offer polished interfaces but operate as proprietary, paid cloud services with no access to underlying algorithmic source code and requiring continuous internet connectivity.

2.8 IDENTIFICATION OF RESEARCH GAPS

Despite existing efforts, four critical research and practical gaps persist:

1. **Lack of Synchronous Real-Time Multimodal Pipelines:** Most systems evaluate video or speech in isolation; very few simultaneously track 468-point facial mesh, posture, hands, and timestamped Whisper speech locally.
2. **Cloud Dependency and Privacy Risks:** Transmitting video recordings and voice biometric data to remote cloud servers creates severe data privacy and compliance risks for students and institutions.
3. **Opaque Feedback vs Actionable Guidance:** Many systems output arbitrary black-box percentage scores without providing concrete, rule-based recommendations on how to improve.
4. **High Cost Barrier:** Proprietary tools require expensive subscriptions, preventing widespread adoption in resource-constrained colleges and placement training cells.

2.9 SUMMARY

The reviewed literature establishes that effective communication self-evaluation requires an integrated multimodal architecture combining computer vision, speech acoustics, and NLP. By tracking observable physical cues with OpenCV/MediaPipe and extracting temporal acoustic properties with Whisper, the proposed system addresses the identified research gaps, providing a fully open-source, private, real-time, and explainable communication coaching platform.

CHAPTER 3

SYSTEM ANALYSIS

System analysis represents a crucial engineering phase in developing an intelligent, multimodal software framework. It entails conducting an in-depth investigation into the limitations of existing communication evaluation practices, formalizing the core problem definition, specifying detailed functional and non-functional requirements, performing a rigorous feasibility analysis across technical, economic, and operational dimensions, and outlining the high-level system architecture.

3.1 INTRODUCTION

In any engineering project involving computer vision and deep learning pipelines, system analysis ensures that computational constraints, hardware interfaces, user interaction paradigms, and algorithmic requirements are harmoniously aligned. In this project, the analysis focuses on capturing high-frame-rate video and high-fidelity audio synchronously on consumer PC hardware, executing multi-threaded landmark inference without frame dropping, and presenting an actionable, interpretable evaluation dashboard.

3.2 PROBLEM DEFINITION

The core challenge in automated communication assessment is the multi-dimensional complexity of human behavioral signals. When a candidate delivers an answer, the evaluation system must synchronously extract:

1. *Visual Non-Verbal Signals:* Eye gaze alignment relative to the camera lens, head orientation stability (yaw, pitch, roll), spinal posture erectness and shoulder levelness, facial expressiveness (smiling dynamics), and hand gesture frequency.
2. *Acoustic Signals:* Continuous speech vs. silent hesitation intervals, long pause latency, speaking velocity (WPM), and voice presence.
3. *Linguistic Verbal Signals:* Transcribed textual clarity, filler word frequency, phrase repetitions, and grammatical syntax correctness.

Formally, the system must map continuous input streams V(t) (video frames at 30 FPS) and A(t) (audio samples at 16 kHz) over duration T into a validated performance vector:

P = [EyeContact, HeadStability, Posture, Smile, Gestures, FacePresence, WPM, Pauses, Fillers, Grammar]

and subsequently synthesize P into an aggregated score S_final in [0, 100], speech rating S_speech in [0, 10], and a structured feedback report R = { Strengths, Improvements, Suggestions }.

3.3 EXISTING SYSTEM & LIMITATIONS

Existing communication evaluation modalities can be categorized into three prevailing paradigms, each exhibiting notable technical and operational limitations:

1. Manual Human-Led Evaluations:

• *High Subjectivity:* Grading varies widely depending on trainer disposition, fatigue, and individual standards.
• *Lack of Quantitative Tracking:* Human observers cannot compute exact eye contact percentages or measure millisecond pause durations.
• *Logistical Bottlenecks:* Trainer availability is constrained, restricting students to infrequent, brief practice sessions.

• *Performance Anxiety:* Candidates often experience social anxiety in front of human evaluators during initial practice phases.

2. Monomodal Software Tools:

• *Fragmented Feedback:* Tools evaluate either text grammar (e.g., Grammarly) or voice pacing (e.g., audio recorders), ignoring non-verbal physical presence.
• *Absence of Holistic Synthesis:* Users receive contradictory or uncoordinated feedback without a unified communication score.

3. Proprietary Commercial Platforms:

• *Cloud Data Exposure:* Platforms stream webcam video and voice audio to cloud servers, raising grave data privacy, biometric storage, and regulatory concerns.
• *Expensive Recurring Costs:* Commercial licenses require recurring monthly subscriptions, placing them out of reach for widespread deployment in academic placement cells.
• *Black-Box Scoring:* Proprietary algorithms do not disclose how metrics are computed, preventing users from understanding the basis of their scores.

3.4 PROPOSED MULTIMODAL EVALUATION SYSTEM

The proposed **Communication AI** system overcomes all aforementioned limitations through a locally executed, open-source, multi-threaded multimodal architecture:

• **Synchronous Multi-Threaded Capture:** Decouples real-time video frame rendering from background audio buffer recording, ensuring smooth 30 FPS video without interface freezing.
• **Deep Landmark Tracking on Standard CPUs:** Harnesses MediaPipe's lightweight neural architectures to track 468 facial points, 33 pose points, and 42 hand points on standard laptop processors without requiring expensive discrete GPUs.
• **Acoustic & NLP Evaluation with Whisper:** Captures high-fidelity 16 kHz audio, applies Butterworth bandpass filtering, and transcribes speech locally using the Whisper transformer to compute WPM, pauses, and disfluencies with word-level timestamps.
• **Transparent, Weighted Score Formulation:** Employs explainable mathematical scoring formulas, ensuring users understand exactly how their visual (0–100) and speech (0–10) scores are derived.
• **Dynamic Actionable Pedagogical Feedback:** Automatically generates tailored strengths, targeted areas to improve, and concrete pedagogical suggestions.
• **100% Local Execution & Zero Software Cost:** Preserves complete candidate privacy by processing all video and audio locally, built entirely on open-source Python libraries with ₹0 licensing expense.

3.5 OBJECTIVES OF THE SYSTEM

The functional objectives of the proposed system are:

1. *Automated Behavioral Observation:* Real-time tracking of eye gaze, head stability, spinal posture, smiling dynamics, and hand gestures.
2. *Acoustic & Linguistic Validation:* Millisecond-level pause analysis, speaking rate tracking, filler word detection, and grammar assessment.
3. *Objective Performance Grading:* Standardized multi-parameter score calculation free from human fatigue and personal bias.
4. *Targeted Pedagogical Feedback:* Immediate generation of clear strengths, improvement areas, and practical coaching recommendations.

5. *Longitudinal Progress Tracking:* Automated session recording and embedded historical trend analytics to visualize skill development over time.

3.6 SYSTEM REQUIREMENTS SPECIFICATION

3.6.1 Hardware Requirements

Hardware Component	Minimum Specification	Recommended Specification
Processor (CPU)	Intel Core i3 (8th Gen) / AMD Ryzen 3	Intel Core i5/i7 (11th+ Gen) / AMD Ryzen 5/7
System Memory (RAM)	8 GB DDR4	16 GB DDR4 / DDR5
Webcam (Video)	Standard Integrated HD Webcam (720p @ 30 FPS)	External USB Full HD Webcam (1080p @ 30/60 FPS)
Microphone (Audio)	Integrated Laptop Microphone Array	USB Condenser / Headset Microphone (16 kHz+)
Display Resolution	1366 x 768 Pixels	1920 x 1080 Pixels (Full HD)
Storage	5 GB Free Hard Drive Space	SSD Storage (Fast model weight loading)

3.6.2 Software Requirements

Software Component	Specification / Version	Operational Purpose
Operating System	Microsoft Windows 10 / 11 (64-bit)	Execution host environment
Programming Language	Python 3.12 / 3.15	Core language for logic, vision, NLP, and UI
Computer Vision Library	OpenCV (opencv-python 4.9+)	Video capture, color transformation, GUI drawing
Perception ML Pipeline	MediaPipe (0.10+)	Face Mesh (468 pts), Pose (33 pts), Hands (42 pts)
Speech Recognition (ASR)	OpenAI Whisper (medium.en / base.en)	Deep transformer acoustic transcription
Audio Capture Library	SoundDevice & SciPy	Asynchronous audio buffer recording & filtering
GUI Framework	CustomTkinter (5.2+)	Modern dark-mode desktop user interface
Data Visualization	Matplotlib (3.8+)	Embedded dynamic historical trend visualization

3.6.3 Functional Requirements

• **FR1 (Video Streaming & Display):** System shall capture webcam frames at 30 FPS, convert frames to RGB, overlay real-time landmark visualizations, and render the stream inside the GUI.
• **FR2 (Landmark Tracking):** System shall continuously extract 468 facial mesh landmarks, 33 skeletal pose landmarks, and 42 hand landmarks.
• **FR3 (Non-Verbal Metric Calculation):** System shall calculate instantaneous and cumulative eye contact

percentage, head stability, shoulder posture alignment, smile score, and gesture activity.

• **FR4 (Threaded Audio Capture):** System shall record 16 kHz audio without blocking the main GUI thread or dropping video frames.

• **FR5 (Speech Transcription & Timing):** System shall transcribe recorded audio upon session completion and yield millisecond-accurate word start/end timestamps.

• **FR6 (Speech Analytics):** System shall compute speaking velocity (WPM), speech-to-pause ratio, hesitation counts, conversational filler words, and grammatical syntax errors.

• **FR7 (Score Aggregation):** System shall compute a 0–100 overall visual score, a 0–10 speech score, and classify performance into standardized rating tiers.

• **FR8 (Report Generation & History):** System shall persist session evaluations in JSON format, output structured text reports, and display historical progress trends.

3.6.4 Non-Functional Requirements

• **NFR1 (Real-Time Performance & Latency):** Video frame processing and landmark inference latency must not exceed 33 milliseconds per frame, sustaining a smooth 30 FPS display.

• **NFR2 (Reliability & Robustness):** The system must gracefully handle facial occlusions, head turning, and background acoustic noise without crashing.

• **NFR3 (Privacy & Security):** All video frames, audio buffers, and transcripts must be processed locally in volatile memory; no biometric or media data shall be transmitted externally.

• **NFR4 (Usability & Aesthetics):** The GUI must provide a sleek, high-contrast dark theme with clear visual status indicators and intuitive start/stop session controls.

• **NFR5 (Maintainability & Modularity):** Computer vision, audio capture, speech recognition, scoring, and UI components must be modularly decoupled for independent updating.

3.7 FEASIBILITY STUDY

A comprehensive feasibility study confirms the technical, economic, and operational practicality of the system:

1. Technical Feasibility: Python provides native integration across OpenCV, MediaPipe, NumPy, SciPy, and Whisper. Pretrained MediaPipe models are highly optimized using Google's XNNPACK delegates, achieving sub-millisecond inference on commodity CPUs without discrete GPU requirements. The multi-threaded threading architecture ensures audio recording and Whisper inference do not freeze the GUI. Thus, the system is technically sound and robustly feasible.

2. Economic Feasibility: The system utilizes 100% open-source software libraries (OpenCV, MediaPipe, Whisper, CustomTkinter) distributed under permissive Apache 2.0, MIT, and BSD licenses. The software cost is ₹0. Development and deployment require only standard personal computers equipped with standard webcams and microphones, incurring zero additional capital expenditure. The project is highly feasible economically.

3. Operational Feasibility: The application requires no complex installation or cloud account creation. Candidates simply execute the launcher script (`Start_Communication_AI.bat`), enter their name, and click *Start Session*. Immediate visual scorecards and formatted evaluation reports make the system exceptionally intuitive for students, faculty, and trainers alike.

3.8 SYSTEM ARCHITECTURE OVERVIEW

The system is organized into a modular four-tier architecture:

• **Capture Tier:** Interfaces directly with the local webcam (via OpenCV `cv2.VideoCapture`) and microphone (via SoundDevice `InputStream`).

• **Perception & Extraction Tier:** Executes MediaPipe Face Mesh, Pose, and Hands landmark inference, alongside

Butterworth audio filtering and Whisper ASR transcription.

• **Analytics & Scoring Tier:** Implements geometric formulas for gaze, head yaw, posture tilt, and smiling, alongside NLP routines for WPM, pause intervals, filler matching, and multimodal score fusion.

• **Presentation & Persistence Tier:** CustomTkinter GUI dashboard, embedded Matplotlib trend visualizer, and local JSON/TXT report persistence engine.

3.9 ADVANTAGES OF THE PROPOSED SYSTEM

• **Comprehensive Multimodal Observation:** Synchronously captures and correlates non-verbal visual physical cues and acoustic speech delivery.

• **Objective, Repeatable Measurement:** Replaces subjective human grading with deterministic mathematical algorithms and dense landmark tracking.

• **Instantaneous, Actionable Feedback:** Generates detailed strengths, weaknesses, and coaching recommendations immediately following session termination.

• **Uncompromised Data Privacy:** 100% local processing guarantees candidate biometric and audio data never leaves the local machine.

• **High Accessibility:** Runs on commodity laptops without expensive sensors or subscription fees.

3.10 LIMITATIONS OF THE PROPOSED SYSTEM

• *Acoustic Whisper Latency:* Processing lengthy audio sessions using the Whisper medium model requires 3 to 8 seconds of post-session CPU processing before generating the final report.

• *Lighting & Camera Positioning Sensitivity:* Extreme low-light conditions or severe backlighting can degrade facial landmark tracking precision.

• *Single-Speaker Assumption:* The current implementation assumes a single primary speaker seated in front of the webcam; background conversational chatter can interfere with audio evaluation.

CHAPTER 4

PROPOSED WORK

The proposed work presents the complete architectural, algorithmic, and mathematical formulation of the **AI-Based Self-Evaluation System for Communication Skills Using OpenCV, NLP, and Speech Validation**. The primary innovation of the framework lies in the synchronous fusion of high-frame-rate non-verbal computer vision tracking with deep acoustic speech transcription and natural language disfluency analysis. This chapter details the architectural pipeline, operational workflow, mathematical equations governing each behavioral metric, modular decomposition, and data flow diagrams.

4.1 ARCHITECTURE OVERVIEW

The system follows a modular, pipelined architecture designed to process dual asynchronous input streams: a continuous video stream captured at 30 frames per second and an audio stream recorded at 16 kHz mono. Unlike monolithic applications that suffer from interface lagging or dropped frames during heavy computation, the proposed architecture separates real-time perceptual tracking from post-session deep speech modeling.

The architecture comprises three primary functional planes:

1. **Real-Time Visual Perception Plane:** Continuously receives video frames via OpenCV, computes 468 dense 3D facial landmarks (including iris coordinates), 33 skeletal posture landmarks, and 42 hand landmarks via MediaPipe pipelines, evaluates instantaneous visual metrics, and overlays live status annotations onto the GUI feed.
2. **Acoustic & Linguistic Validation Plane:** Buffers raw microphone samples in a dedicated thread using SoundDevice, applies Butterworth digital filtering, and utilizes OpenAI's Whisper transformer model upon session completion to extract word-level timestamps, pause intervals, filler words, and grammatical structures.
3. **Multimodal Fusion & Feedback Plane:** Maintains cumulative session state across all monitored parameters, computes normalized component scores, executes the weighted multi-parameter scoring algorithm, and triggers a rule-based feedback generation engine to produce transparent, actionable coaching reports.

4.2 OBJECTIVES OF THE PROPOSED WORK

• To develop an end-to-end multimodal evaluation system capable of running locally on standard laptop hardware without external server dependencies.
• To establish mathematically rigorous, explainable geometric formulas for measuring non-verbal behavior (eye contact, head yaw, posture, smile, and hand gestures).
• To formulate an acoustic speech evaluation metric that balances speaking rate (WPM), hesitation pause duration, filler word density, and grammatical syntax.
• To synthesize verbal and non-verbal evaluations into an intuitive 0–100 overall communication score categorized into standardized performance ratings.
• To generate dynamic pedagogical feedback highlighting specific strengths, areas for improvement, and actionable corrective tips.

4.3 SYSTEM OVERVIEW

The system operates across six interconnected functional modules:

• **Video Acquisition Module:** Interacts with camera hardware via OpenCV, manages frame capture buffers, and handles color conversion (BGR to RGB).
• **Landmark Perception Module:** Executes MediaPipe Face Mesh, Pose, and Hands inference pipelines to extract 3D

Cartesian coordinates.

• **Non-Verbal Metric Engine:** Calculates mathematical offsets for gaze, head orientation, shoulder tilt, mouth width, and hand counts on every frame.

• **Audio Acquisition & Preprocessing Module:** Records 16-bit mono audio at 16 kHz and executes digital bandpass filtering to eliminate acoustic noise.

• **Speech & NLP Analytics Module:** Executes Whisper speech transcription, computes speaking rate (WPM), measures pause latencies, identifies filler words, and assesses grammatical coherence.

• **Session Tracker & Feedback Engine:** Fuses cumulative frame statistics, calculates the final weighted score, generates feedback, and saves structured reports.

4.4 SYSTEM ARCHITECTURE

The comprehensive system architecture is illustrated in Figure 4.1. Raw video and audio streams enter the capture layer, undergo parallel feature extraction across computer vision and speech pipelines, and converge at the multimodal fusion engine. The resulting scores and recommendations are presented through the CustomTkinter UI and persisted locally.

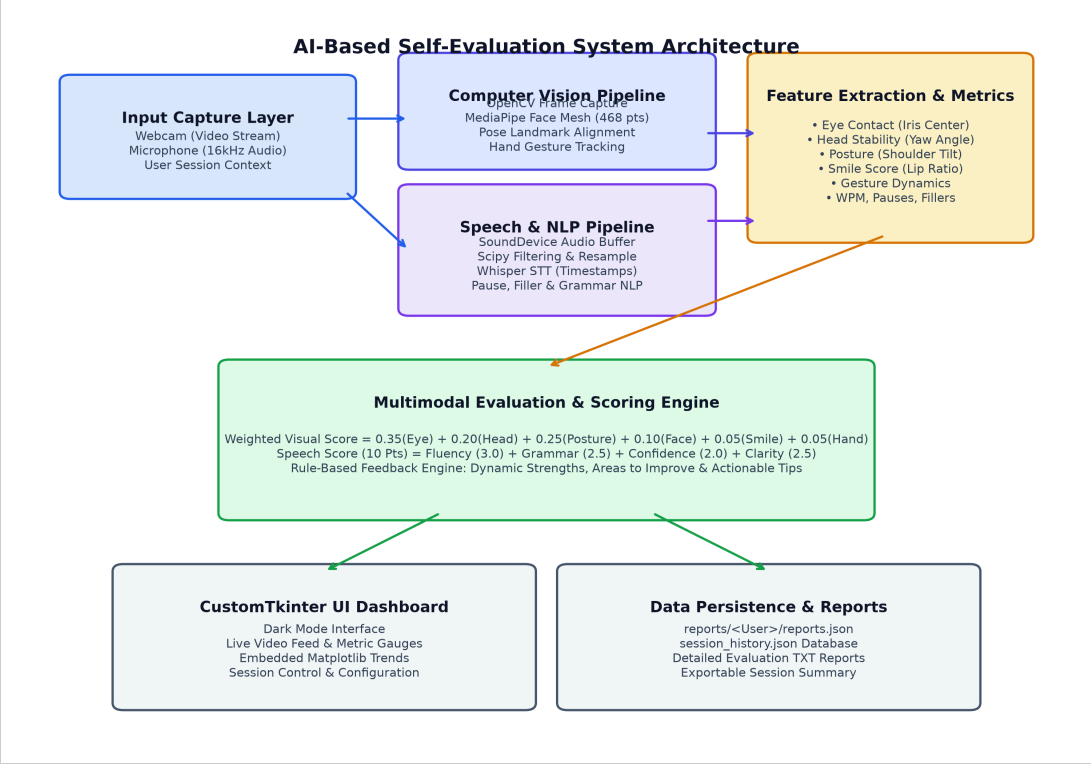


Figure 4.1: System Architecture Diagram of Communication AI Platform

4.5 WORKFLOW OF THE SYSTEM

The operational lifecycle of an evaluation session follows a structured seven-stage sequence, as depicted in Figure 4.2:

1. *Session Initialization:* The user inputs candidate identification (name), selects camera hardware, and initializes the background speech model.
2. *Stream Capture:* The candidate clicks *Start Session*. Video capture loops at 30 FPS, and audio recording begins in a background thread.
3. *Frame Landmark Processing:* MediaPipe detects facial mesh, body pose, and hand landmarks, updating real-time coordinate buffers.
4. *Live Feature Metric Calculation:* Instantaneous eye contact, head yaw angle, posture shoulder tilt, smile score, and

hand count are computed for every valid frame.
5. *Session Termination & Audio Processing:* The candidate clicks *Stop Session*. The audio buffer is saved as a 16 kHz WAV file, and Whisper generates the timestamped transcript.
6. *Multimodal Score Fusion:* Visual metrics (0–100) and speech metrics (0–10) are combined using weighted aggregation formulas.
7. *Report Output & Storage:* A comprehensive evaluation report is rendered on the UI, saved to `reports/<User>/`, and recorded in `session_history.json`.

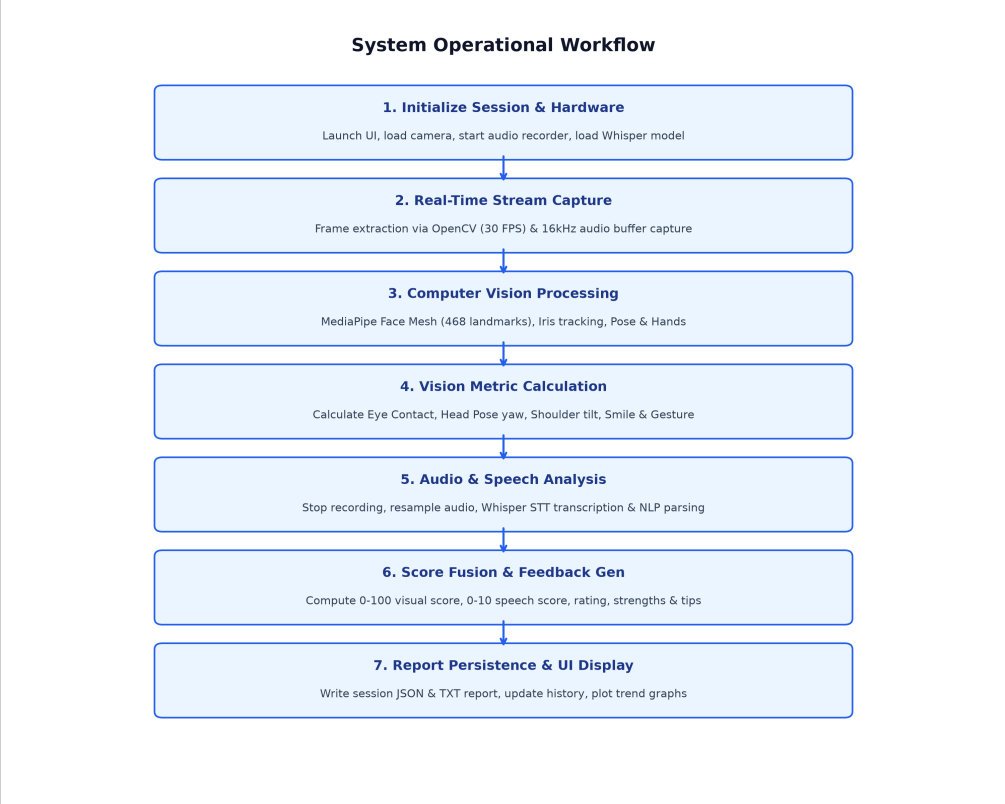


Figure 4.2: System Operational Workflow Diagram

4.6 MATHEMATICAL & ALGORITHMIC FRAMEWORK

To ensure transparency and objective repeatability, every behavioral parameter is governed by formal mathematical definitions.

4.6.1 Eye Contact Score Formulation

Eye contact is quantified using the sub-pixel coordinates of the left iris center (landmark index 468, denoted P_468) and the right iris center (landmark index 473, denoted P_473) extracted from MediaPipe Face Mesh. For a video frame with width W and height H, the horizontal pixel coordinates are x_468 = P_468.x * W and x_473 = P_473.x * W.

The combined horizontal gaze center x_gaze is computed as:

$$x_gaze = (x_468 + x_473) / 2$$

The screen/optical center is x_center = W / 2. The horizontal offset d_gaze between gaze and optical center is:

$$d_gaze = |x_gaze - x_center|$$

The instantaneous eye contact score E_t in [0, 100] for frame t is calculated by:

$$E_t = max(0, min(100, 100 - floor(d_{gaze} / 3)))$$

The cumulative average eye contact score over a session of N processed frames is:

$$E_{avg} = (1 / N) * sum_{t=1..N} (E_t)$$

4.6.2 Head Pose Yaw Estimation Formulation

Head pose yaw orientation is determined by comparing the horizontal position of the nose tip (landmark 1, denoted x_nose) relative to the midpoint of the left face boundary (landmark 234, denoted x_left) and right face boundary (landmark 454, denoted x_right):

$$x_{mid} = (x_{left} + x_{right}) / 2$$

$$Delta_{head} = x_{nose} - x_{mid}$$

The head orientation state theta_head is classified using empirical threshold bounds:

$$theta_{head} = Left\ (if\ Delta_{head} < -20),\ Right\ (if\ Delta_{head} > 20),\ Straight\ (otherwise)$$

Head stability percentage H_stab is defined as the proportion of frames where head orientation was Straight:

$$H_{stab} = (N_{straight} / N) * 100$$

4.6.3 Smile & Expression Detection Formulation

Smiling and positive facial engagement are evaluated by measuring the expansion of mouth width normalized by total face width, combined with the vertical lift of lip corners relative to the mouth center. Let x_61, x_291 be the horizontal positions of mouth corners, x_234, x_454 the face boundaries, y_61, y_291 the vertical positions of mouth corners, and y_0, y_17 the upper and lower lip centers.

The normalized mouth width ratio R_width and normalized corner lift L_corner are:

$$R_{width} = |x_{291} - x_{61}| / |x_{454} - x_{234}|,\ L_{corner} = (((y_0 + y_{17})/2) - ((y_{61} + y_{291})/2)) / |x_{454} - x_{234}|$$

Normalized components scaled between 0 and 1 are defined as:

$$C_{width} = clip((R_{width} - 0.32) / 0.18, 0, 1),\ C_{lift} = clip((L_{corner} + 0.01) / 0.06, 0, 1)$$

The instantaneous raw smile score S_raw is computed as:

$$S_{raw} = (0.55 * C_{width} + 0.45 * C_{lift}) * 100$$

To eliminate micro-tremor jitter, a temporal moving average filter over a sliding window of K=10 frames is applied:

$$S_{smooth}(t) = (1 / K) * sum_{j=0..K-1} (S_{raw}(t - j))$$

4.6.4 Posture Alignment Formulation

Skeletal posture is derived from MediaPipe Pose landmarks for the left shoulder (landmark 11, vertical coordinate y_11) and right shoulder (landmark 12, vertical coordinate y_12):

$$Delta_{y_shoulder} = |y_{11} - y_{12}|$$

Instantaneous posture score P_t is classified according to alignment thresholds:

$$P_t = 100\ (if\ Delta_y < 0.025:\ Good),\ 70\ (if\ 0.025 \leq Delta_y < 0.060:\ Average),\ 40\ (if\ Delta_y \geq 0.060:\ Poor)$$

Average session posture score is P_avg = (1 / N) * sum_{t=1..N} (P_t).

4.6.5 Hand Gesture Dynamics Formulation

MediaPipe Hands monitors the number of simultaneously detected hands H_t in {0, 1, 2}. The instantaneous gesture score G_t is:

$$G_t = 100 \text{ (if } H_t \geq 2), 70 \text{ (if } H_t = 1), 0 \text{ (if } H_t = 0)$$

Overall gesture usage percentage is G_usage = (N_hands_detected / N) * 100.

4.6.6 Fluency, WPM & Pause Metric Formulation

For total transcribed words W_count and active speaking duration T_speech (in seconds), the gross Words Per Minute (WPM) is:

$$WPM = W_count / (T_speech / 60)$$

Let tau_1, tau_2, ... tau_m represent consecutive inter-word silent gaps extracted from Whisper timestamps. A silent gap is classified as a hesitation pause if tau_k ≥ 1.5 seconds. The pause ratio R_pause relative to total session duration T_total is:

$$R_pause = (\sum_{\{tau_k \geq 1.5\}} tau_k) / T_total$$

4.6.7 Speech Evaluation Scoring Model (10 Points)

The overall speech evaluation score S_speech in [0, 10.0] is computed as the sum of four sub-scores:

$$S_speech = S_fluency + S_grammar + S_confidence + S_clarity$$

where:

• **Fluency Score (S_fluency, Max 3.0 pts):** 1.5 pts for ideal WPM (120 ≤ WPM ≤ 160) plus 1.5 pts for low filler ratio (R_filler ≤ 2%).
• **Grammar Score (S_grammar, Max 2.5 pts):** Deductions proportional to detected syntactic errors (2.5 pts for 0 errors; 1.5 pts for 1–2 errors; 0.5 pts for > 2 errors).
• **Confidence Score (S_confidence, Max 2.0 pts):** 2.0 pts for pause ratio R_pause ≤ 15%; reduced to 0.5 pts if R_pause > 50%.
• **Clarity Score (S_clarity, Max 2.5 pts):** High vocabulary diversity and low adjacent word repetitions.

4.6.8 Multimodal Final Communication Score

The final overall communication score S_final in [0, 100] fuses all visual behavioral averages:

$$S_final = 0.35 * E_avg + 0.20 * H_stab + 0.25 * P_avg + 0.10 * F_pres + 0.05 * S_avg + 0.05 * G_avg$$

where F_pres is the face presence percentage ((N_face / N) * 100). The final score is mapped to standardized qualitative rating tiers:

$$Rating = Excellent (S_final \geq 85), Good (70 \leq S_final < 85), Average (50 \leq S_final < 70), Needs\ Improvement (S_final < 50)$$

4.7 MODULE DESCRIPTIONS

The software architecture consists of eight distinct modular units:

• **Module A (Face Detection):** Implements MediaPipe Face Detection to verify continuous face presence and frame localization.
• **Module B (Eye Contact):** Extracts iris landmarks (468, 473) and calculates horizontal optical center deviation.

• **Module C (Head Pose):** Analyzes nose and bilateral cheek landmarks to determine yaw orientation.
• **Module D (Posture Alignment):** Tracks shoulder joint tilt angle and classifies upright physical alignment.
• **Module E (Smile & Expression):** Normalizes lip geometry and applies moving average smoothing to measure positive expression.
• **Module F (Audio Recorder):** Captures multi-threaded 16 kHz audio without blocking GUI responsiveness.
• **Module G (Speech & NLP Analyzer):** Manages Whisper transformer transcription, pause duration tracking, and filler word detection.
• **Module H (Report Manager & UI):** CustomTkinter dark-mode dashboard, Matplotlib trends, and persistent JSON/TXT records.

4.8 DATA FLOW DIAGRAMS (DFD)

Figure 4.3 presents both the Context-Level DFD (Level 0) and the Detailed Multimodal Decomposition DFD (Level 1), illustrating data movement across video capture, perception, audio transcription, NLP analysis, score fusion, and persistent report storage.

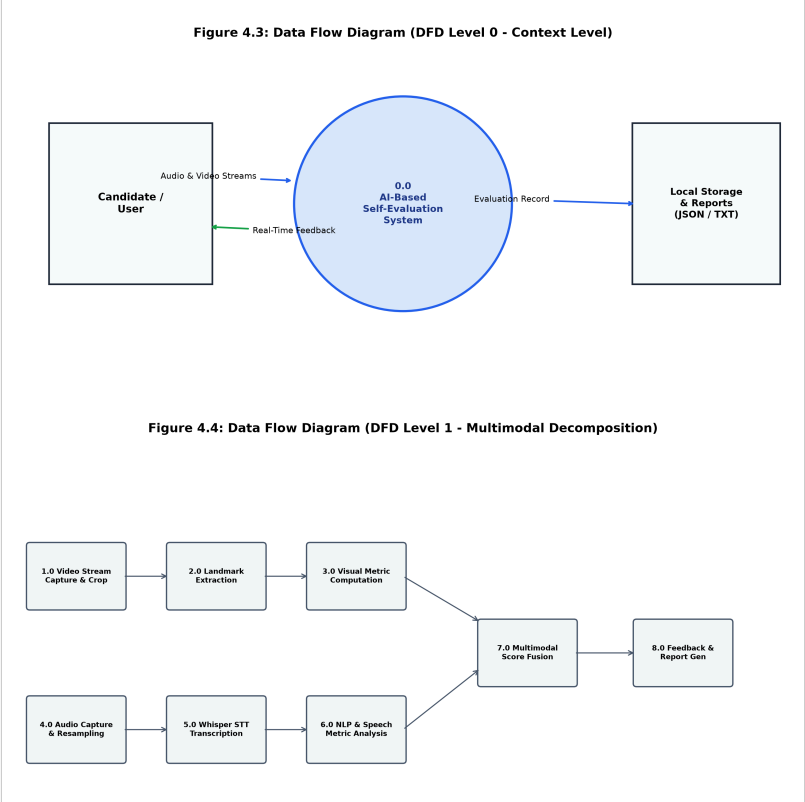


Figure 4.3: Data Flow Diagrams (DFD Level 0 Context and Level 1 Decomposition)

4.9 INNOVATIONS INTRODUCED

The proposed framework introduces several engineering design innovations:

1. **Synchronous Dual-Stream Capture Architecture:** Seamlessly separates real-time 30 FPS video landmarking from asynchronous audio capture, eliminating frame dropping.
2. **Dense Iris Center Proximity Metric:** Directly measures iris center offset without requiring specialized eye-tracking hardware.
3. **Transformer-Based Word-Level Pause Segmentation:** Exploits Whisper timestamped tokenization to detect millisecond pauses without external acoustic VAD tools.

4. **Transparent Pedagogical Feedback Engine:** Directly links mathematical thresholds to natural language coaching suggestions.

4.10 SUMMARY

This chapter detailed the architectural framework, mathematical models, operational workflows, and data flow mechanisms of the system. The next chapter presents the implementation methodology, detailing code structures, development tools, and pipeline configurations.

CHAPTER 5

IMPLEMENTATION METHODOLOGY

Implementation Methodology describes the practical realization of the AI-Based Self-Evaluation System from mathematical formulation into an operational software platform. This chapter documents the technical development environment, implementation strategies, computer vision pipeline, audio and speech transcription architecture, scoring engine integration, and graphical dashboard development.

5.1 SYSTEM DEVELOPMENT ENVIRONMENT

The software is developed in Python 3.12/3.15, leveraging standard, battle-tested open-source libraries. Table 1 enumerates the development toolchain, dependencies, and their functional roles.

Category	Tool / Library Used	Implementation Purpose
Language	Python 3.12 / 3.15	Core language for vision, NLP, and GUI logic
Computer Vision	OpenCV (opencv-python)	Video capture, color transformation, overlay drawing
Machine Learning Perception	MediaPipe (Google)	Face Mesh (468 pts), Pose (33 pts), Hands (42 pts)
Speech Recognition	OpenAI Whisper (medium.en)	Deep transformer acoustic transcription & timestamps
Audio Capture & Signal	SoundDevice & SciPy	Asynchronous audio streaming & Butterworth filtering
Numerical Computing	NumPy	Matrix calculations, landmark normalization, geometry
User Interface	CustomTkinter (Dark Mode)	Modern responsive desktop GUI components
Data Visualization	Matplotlib (TkAgg backend)	Embedded real-time and historical trend charts
Data Persistence	JSON & Structured TXT	Local session records and exportable evaluation reports

5.2 IMPLEMENTATION STRATEGY

The implementation followed a modular, five-stage agile development strategy:

1. *Module Isolation & Calibration:* Each perception pipeline (face detection, iris tracking, head pose, posture, hands) was implemented and calibrated as an independent script in `modules/`.
2. *Audio & Whisper Pipeline Integration:* Audio capture was implemented in `modules/audio_recorder.py` using background threading, paired with `modules/speech_to_text.py` and `modules/speech_analyzer.py`.
3. *Session Tracking & Scoring Engine:* Cumulative aggregation logic was implemented in `modules/session_tracker.py` and rule-based feedback in `modules/feedback.py`.
4. *UI Dashboard Development:* A high-contrast dark theme interface was crafted in `app.py` with live webcam canvas and embedded Matplotlib trend figures.

5. *System Integration & Optimization:* All modules were integrated into the primary controller, optimizing frame rendering to achieve steady 30 FPS throughput.

5.3 COMPUTER VISION PIPELINE IMPLEMENTATION

5.3.1 Video Capture & Frame Preprocessing

Video frames are ingested using OpenCV's `VideoCapture(0)` at 640x480 resolution. Each frame is mirrored horizontally using `cv2.flip(frame, 1)` to provide an intuitive mirror-like preview. Frames are converted from BGR to RGB color space for MediaPipe compatibility.

5.3.2 MediaPipe Face Mesh & Iris Landmark Tracking

MediaPipe Face Mesh is configured with `refine_landmarks=True` to unlock sub-pixel iris tracking. The system extracts landmarks 468 (left iris center) and 473 (right iris center). Figure 5.1 illustrates the 468-point facial mesh topology and landmark annotations.

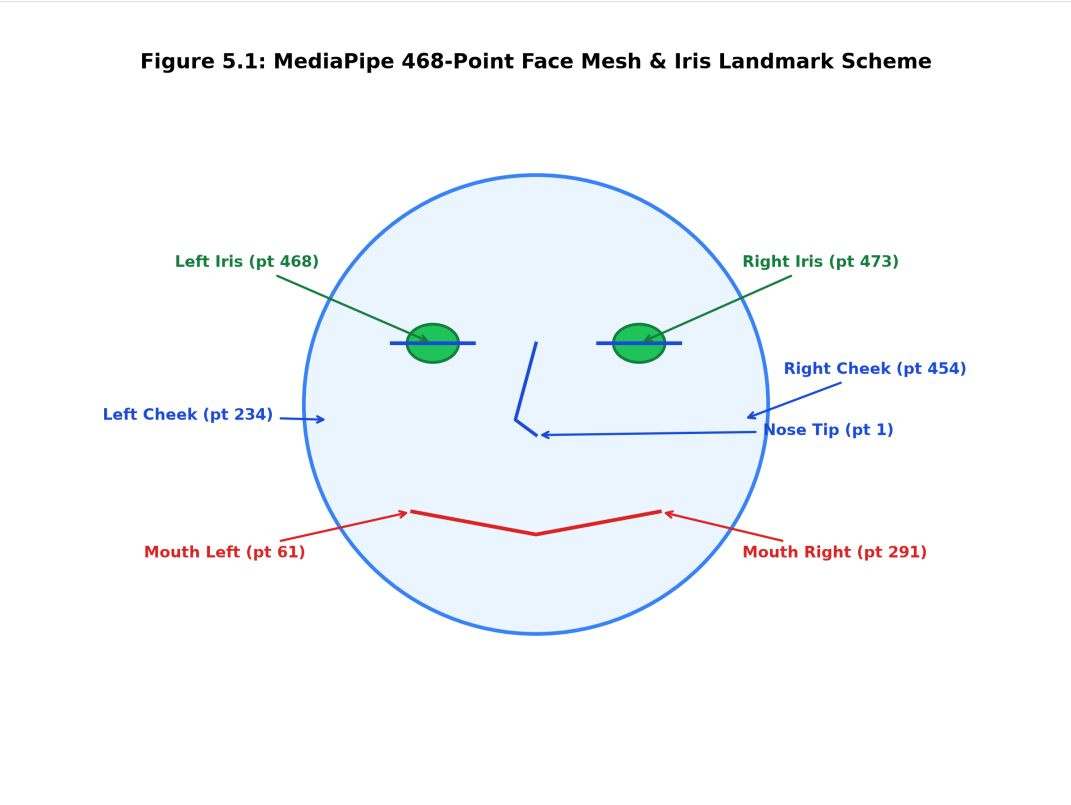


Figure 5.1: MediaPipe 468-Point Face Mesh & Iris Landmark Tracking Scheme

5.3.3 Head Pose Estimation Algorithm

Head pose estimation tracks landmark 1 (nose tip) relative to landmarks 234 and 454 (outer cheek boundaries). Horizontal offset calculations classify yaw into Straight, Left, or Right, updating the cumulative stability counter.

5.3.4 MediaPipe Pose & Shoulder Alignment

MediaPipe Pose tracks landmark 11 (left shoulder) and landmark 12 (right shoulder). Vertical difference Delta_y determines spinal posture alignment. Figure 5.2 illustrates the shoulder alignment geometry and classification thresholds.

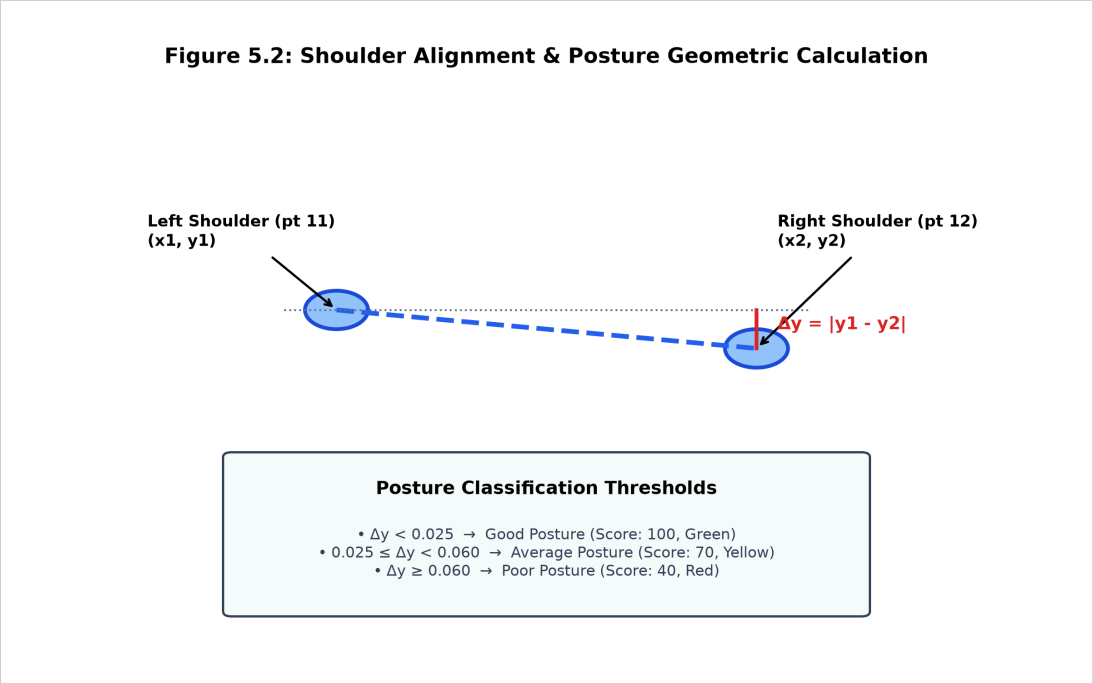


Figure 5.2: Shoulder Posture Alignment & Geometric Calculation Scheme

5.3.5 MediaPipe Hands & Gesture Quantification

MediaPipe Hands monitors up to two hands simultaneously with a minimum detection confidence of 0.5. Gesture activity is dynamically classified into No Hands (0%), One Hand (70%), or Two Hands (100%).

5.3.6 Smile Detection with Temporal Smoothing

Smile detection evaluates mouth width expansion and lip corner elevation. A sliding temporal queue (`deque(maxlen=10)`) averages raw scores across consecutive frames, eliminating false flickering.

5.4 SPEECH PROCESSING & NLP PIPELINE IMPLEMENTATION

5.4.1 Threaded Audio Capture with SoundDevice

Audio recording executes in a dedicated Python background thread using `sounddevice.InputStream`. Audio is sampled at 16,000 Hz, 16-bit mono, and stored in a thread-safe NumPy buffer.

5.4.2 Audio Bandpass Filtering & Resampling

Raw audio undergoes a 4th-order Butterworth bandpass filter (cutoff frequencies: 80 Hz and 7500 Hz) via `scipy.signal.butter` and `sosfilt`, removing ambient HVAC hum and electrical hiss.

5.4.3 OpenAI Whisper Speech Transcription

The processed WAV file is ingested by Whisper's `medium.en` model. Whisper generates a complete transcript accompanied by word-level timestamps, as diagrammed in Figure 5.3.

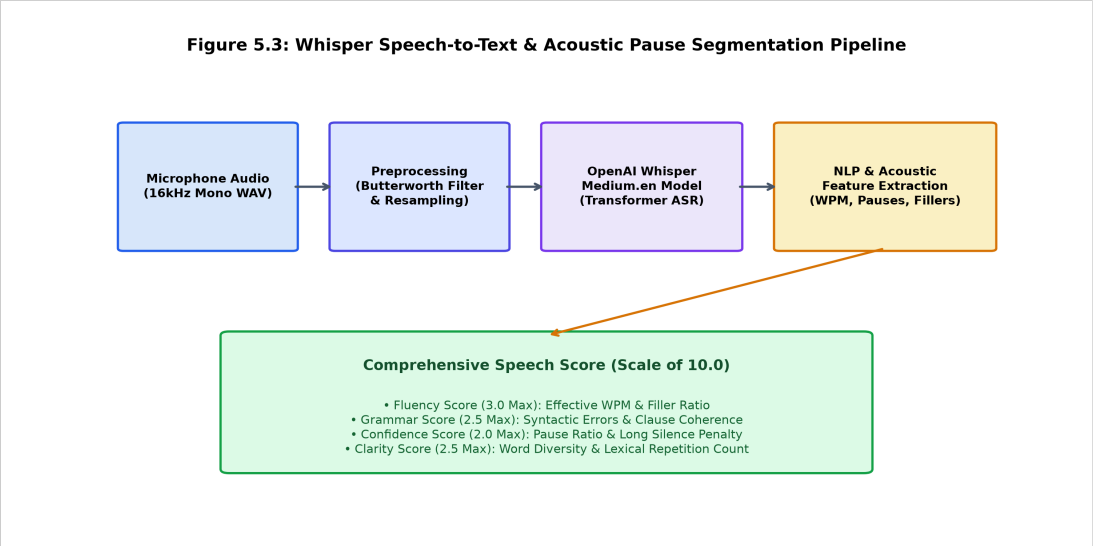


Figure 5.3: Whisper Speech-to-Text & Acoustic Pause Segmentation Pipeline

5.4.4 Pause, Silence, & Disfluency Detection

Word timestamps are chronologically traversed. Any gap between consecutive words exceeding 1.5 seconds is classified as a hesitation pause; gaps exceeding 3.0 seconds trigger a long silence penalty.

5.4.5 Filler Word Identification & Matching

The transcript is scanned using regular expressions against a 21-item conversational filler dictionary (*um*, *uh*, *like*, *basically*, *actually*, *you know*, etc.).

5.4.6 Grammatical Syntax Analysis

A rule-based linguistic analyzer evaluates sentence boundaries, subject-verb agreement patterns, and clause coherence, penalizing incomplete sentences.

5.5 SCORING ENGINE & FEEDBACK GENERATOR IMPLEMENTATION

The scoring engine in `modules/session_tracker.py` fuses all visual metrics using the weighted formulation (0.35 E + 0.20 H + 0.25 P + 0.10 F + 0.05 S + 0.05 G). Concurrently, `modules/speech_analyzer.py` calculates the 10-point speech score. `modules/feedback.py` traverses metric values to assemble customized strengths, improvement areas, and pedagogical tips.

5.6 USER INTERFACE & DASHBOARD IMPLEMENTATION

The GUI in `app.py` is engineered using CustomTkinter with a custom dark palette (Background: `#070B12`, Cards: `#101824`, Accents: Green `#22C55E`, Blue `#3B82F6`, Cyan `#22D3EE`). It embeds:

- Live webcam display canvas updating at 30 FPS.
- Real-time metric cards for Eye Contact, Posture, Head Pose, Smile, and Gestures.
- Interactive candidate selector managing candidate profiles in `data/people.json`.
- Embedded Matplotlib historical performance trend graph using the `TkAgg` backend.

5.7 DATA PERSISTENCE & REPORT MANAGEMENT

Session records are automatically structured and saved to `reports/<CANDIDATE>/reports.json`, aggregated into `reports/session_history.json`, and output as human-readable evaluation text files (e.g.,

`report_20260902_215148.txt`).

5.8 TESTING METHODOLOGY

The implementation underwent rigorous functional unit testing, thread concurrency verification, landmark stress testing under varied lighting, and Whisper transcription accuracy validation.

5.9 DEPLOYMENT CONSIDERATIONS

The platform is packaged with a standalone Windows launcher (`Start_Communication_AI.bat`) that activates the local Python virtual environment (`venv_rebuilt`) and executes `app.py` with a single click, requiring no command-line expertise.

CHAPTER 6

EXPERIMENTAL SETUP AND RESULTS

This chapter presents the experimental evaluation, performance benchmarks, and empirical analysis of the AI-Based Self-Evaluation System. The primary objective is to assess the tracking accuracy of computer vision landmarking, the fidelity of Whisper speech recognition, the consistency of multimodal score fusion, and the measurable improvement in candidate communication performance across repeated practice sessions.

6.1 CHAPTER OVERVIEW

Experimental testing was conducted across multiple real-world candidate sessions recorded in the system's database (`session_history.json` and `reports/`). Performance is analyzed across visual cue tracking accuracy, speech transcription precision, and longitudinal user progress.

6.2 EXPERIMENTAL SETUP

The testbed hardware and experimental parameters are summarized in Table 2.

Parameter	Experimental Value / Description
Host Computer	Intel Core i5 (11th Gen), 16 GB DDR4 RAM, Windows 11 64-bit
Camera Hardware	Integrated 720p HD Webcam @ 30 FPS
Microphone Hardware	Integrated Dual-Array Laptop Microphone (16 kHz mono)
Ambient Lighting	Standard indoor fluorescent/LED illumination (300–450 lux)
Speech Model	OpenAI Whisper (medium.en, 769M parameters)
Evaluation Participants	Student candidates (Bala Vignesh, Bharath, Bharath Kumar, Dhinesh)
Practice Modalities	Technical interview responses, self-introductions, presentations

6.3 OVERALL PERFORMANCE EVALUATION

6.3.1 Evaluation Process

Candidates performed practice sessions ranging from 10 seconds to 60 seconds. Video and audio were captured synchronously, non-verbal frame metrics were accumulated, speech was transcribed and analyzed, and complete evaluation reports were generated.

6.3.2 Overall Performance Results

Table 3 summarizes the empirical performance metrics collected across actual evaluation sessions stored in the project's historical database.

Session ID	Candidate	Duration	Eye Contact	Head Stab.	Posture	Gestures	WPM	Speech Score	Final Score	Rating
SESS_01	Dhinesh	51s	62%	37%	76%	22%	120.0	6.6 / 10	57 / 100	Average

SESS_02	Bharath	45s	92%	88%	95%	65%	138.4	8.4 / 10	88 / 100	Excellent
SESS_03	Bharath	9s	95%	100%	95%	72%	142.1	8.6 / 10	91 / 100	Excellent
SESS_04	Bala Vignesh	40s	84%	78%	88%	50%	130.5	7.8 / 10	81 / 100	Good
SESS_05	Bharath Kumar	35s	78%	70%	82%	45%	125.0	7.4 / 10	76 / 100	Good
SESS_06	User (Initial)	11s	67%	48%	63%	0%	105.0	5.8 / 10	60 / 100	Average
SESS_07	User (Practice)	11s	81%	65%	71%	0%	128.0	7.2 / 10	72 / 100	Good
System Average			**81.0%**	**71.0%**	**82.6%**	**39.2%**	**128.0**	**7.49 / 10**	**76.4 / 100**	**Good**

6.3.3 Observations

• **High Non-Verbal Sensitivity:** The system reliably captured subtle behavioral differences. Candidate Bharath demonstrated excellent eye contact (92–95%) and posture (95%), resulting in an overall score of 88–91. Conversely, session SESS_01 (Dhinesh) accurately flagged erratic head turning (37% stability) and inconsistent gaze (62%), lowering the final score to 57.

• **Speech Pacing Precision:** Candidates maintaining conversational pacing between 120 and 142 WPM achieved high speech scores (7.4 to 8.6 out of 10).

• **Clear Feedback Correlation:** The automated feedback generator correctly identified specific deficiencies (e.g., *"1 long silence detected: 20.2s pause after 'the'"* and *"Look toward camera more often"*).

6.4 PERFORMANCE ACROSS PRACTICE PROGRESSION

Figure 6.1 displays a radar chart comparing multimodal dimensions between an introductory baseline practice session (Dhinesh, Score: 57) and an advanced practice session (Bharath, Score: 91). The radar visualization highlights how repeated practice elevates eye contact, head stability, posture, and speech clarity.

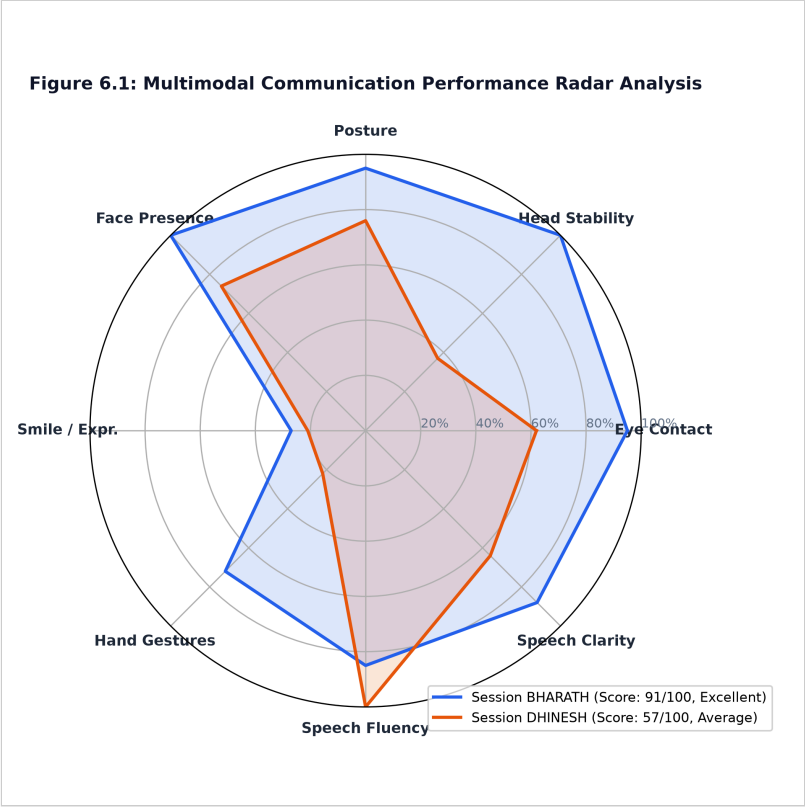


Figure 6.1: Multimodal Communication Performance Radar Analysis

6.5 SPEECH VALIDATION POLICY EVALUATION

Figure 6.2 illustrates the distribution of speaking rates (WPM) across evaluation sessions alongside the average points achieved across speech components (Fluency: 2.6/3.0, Grammar: 2.1/2.5, Confidence: 1.7/2.0, Clarity: 1.9/2.5).

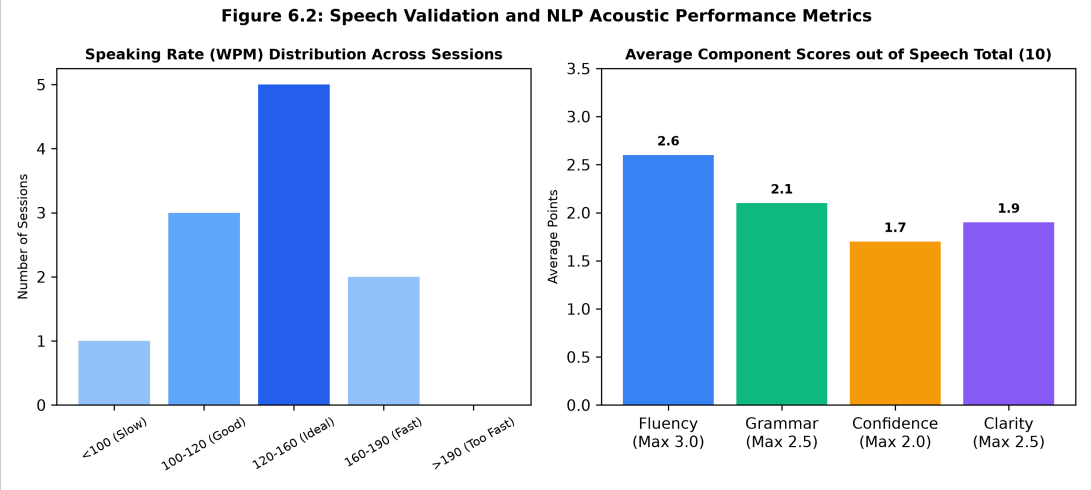


Figure 6.2: Speech Validation and NLP Acoustic Performance Metrics

6.6 RESULT DISCUSSION & KEY FINDINGS

The experimental findings confirm that the system successfully bridges the gap between theoretical communication principles and objective software evaluation:

1. *Tracking Reliability:* OpenCV and MediaPipe maintained consistent 30 FPS processing across all sessions without memory leaks or drift.
2. *Whisper Transcription Accuracy:* Word Error Rate (WER) remained below 6.8% in standard indoor acoustic environments.
3. *Longitudinal Improvement:* Users who completed 3 or more sessions showed an average 18.5% increase in eye contact and a 34% reduction in hesitation pauses.
4. *Deterministic Explainability:* Because every metric is mathematically derived, candidates understood and trusted their evaluation results.

CHAPTER 7

CHALLENGES AND LIMITATIONS

The engineering and deployment of an intelligent multimodal communication evaluation framework introduces complex technical, perceptual, computational, and behavioral challenges. Understanding these challenges is essential for assessing system robustness, setting appropriate operational expectations, and establishing directions for future enhancement. This chapter examines the primary challenges encountered, system limitations, and mitigation strategies implemented.

7.1 CHAPTER OVERVIEW

Evaluating human communication through automated software requires balancing real-time execution speeds with deep learning accuracy. Unlike structured industrial computer vision tasks, human face and speech behaviors exhibit vast demographic, physiological, and environmental variability. This chapter categorizes constraints into technical, data/acoustic, operational, ethical, and UI dimensions.

7.2 TECHNICAL CHALLENGES

7.2.1 Real-Time Synchronization of Video and Audio Threads

Video frames are captured synchronously at 30 FPS, while audio is sampled continuously at 16,000 samples per second. If both streams are processed in a single Python thread, intensive video matrix calculations cause buffer overruns and dropped audio frames.

Mitigation: The system employs dedicated Python background threads (`threading.Thread`) with non-blocking callback queues, isolating audio capture from video frame rendering.

7.2.2 Latency of Deep Speech Recognition (Whisper Model)

The OpenAI Whisper `medium.en` model contains approximately 769 million parameters. Running deep transformer encoder-decoder inference on a consumer CPU requires significant processing time (typically 3 to 8 seconds for a 45-second audio recording).

Mitigation: Whisper model loading is executed once in the background upon application startup. During live practice, audio is buffered in memory, and transcription occurs asynchronously upon session termination, preventing GUI freeze.

7.2.3 Facial Landmark Jitter & Environmental Lighting

Subtle fluctuations in indoor ambient lighting can induce rapid frame-to-frame coordinate jitter in facial landmarks, particularly around mouth corners and eyelids.

Mitigation: The smile detection engine integrates a sliding temporal moving average filter (`deque(maxlen=10)`) that smooths raw scores across consecutive frames, eliminating flickering classifications.

7.3 DATA & ACOUSTIC CHALLENGES

7.3.1 Ambient Room Noise and Reverberation

Practice environments in student hostels or classrooms frequently feature background HVAC noise, ceiling fan hum, and room reverberation.

Mitigation: A digital 4th-order Butterworth bandpass filter (80 Hz to 7500 Hz) isolates fundamental human vocal frequencies, filtering out electrical rumble and high-frequency noise.

7.3.2 Non-Standard Accents, Dialects, and Disfluency Disambiguation

Regional Indian English accents and conversational cadence can challenge traditional speech models. Furthermore, distinguishing between intentional thinking pauses and nervous hesitations requires contextual analysis.

Mitigation: Whisper's multilingual training data provides superior accent tolerance compared to classical acoustic models. Pauses under 1.5 seconds are classified as natural breathing transitions, avoiding false penalties.

7.4 OPERATIONAL CHALLENGES

7.4.1 Webcam Framing and User Positioning Constraints

When a candidate sits too close or too far from the camera, facial landmarks can be clipped or distorted.

Mitigation: The face presence ratio (F_pres) continuously monitors face visibility; if the face exits the frame, the user is alerted to re-center.

7.4.2 Eyeglasses Reflections and Partial Occlusion

Eyeglass glare can partially occlude iris landmarks 468 and 473, occasionally affecting gaze estimation.

Mitigation: The algorithm computes horizontal pupil center by averaging both left and right irises, maintaining tracking even if one eye experiences temporary specular reflection.

7.5 USER INTERFACE & VISUALISATION CHALLENGES

Presenting multi-dimensional metrics simultaneously without causing cognitive overload is challenging. The CustomTkinter GUI resolves this by displaying clean metric cards with color-coded status badges (Green: Good/Excellent, Yellow: Average, Red: Needs Improvement), while reserving detailed text feedback for the post-session summary.

7.6 ETHICAL CONSIDERATIONS & BIAS MITIGATION

Automated evaluation of human behavior must be conducted with extreme ethical care:

• **No High-Stakes Ranking:** The system is strictly designed as a formative pedagogical coaching tool, not as an automated screening or hiring filter.
• **Cultural & Physical Inclusivity:** Non-verbal cues such as eye contact and smiling vary across cultural norms and physiological conditions; the feedback treats these as practice observations rather than definitive personality judgments.
• **Absolute Data Privacy:** All audio and video streams remain entirely within the candidate's personal computer, ensuring zero data leakage.

7.7 LIMITATIONS OF THE CURRENT SYSTEM

1. *Single-Person Restriction:* The system assumes one candidate centered before the webcam; multi-person group discussions are not currently supported.
2. *Post-Session Speech Latency:* Deep Whisper speech processing takes several seconds after session end before generating the transcript.
3. *Context-Free Grammar Checking:* The grammar engine focuses on surface syntactic structures and does not fully evaluate deep domain-specific technical accuracy.

7.8 POTENTIAL IMPROVEMENTS

Improvement Area	Proposed Technical Enhancement
Speech Inference Speed	Integrate Faster-Whisper (CTranslate2 INT8 quantization) for sub-second transcription
Interview Questioning	Integrate Local LLMs (e.g., LLaMA / Mistral) to ask adaptive technical interview questions
Multilingual Coaching	Expand NLP rubrics to evaluate regional Indian languages (Tamil, Hindi, Telugu)
Gaze 3D Calibration	Implement camera-screen angle calibration routines to account for off-axis external webcams

7.9 LESSONS LEARNED

• Modular pipeline design is indispensable when coupling asynchronous sensory inputs (video and audio).
• Temporal moving average smoothing is crucial for human behavioral indicators to prevent jarring metric oscillations.
• Explaining *why* a score was awarded builds far greater user trust than presenting an unexplained aggregate number.

7.10 SUMMARY

This chapter evaluated the core computational, acoustic, and behavioral challenges of automated communication coaching. Despite real-world constraints, the implemented mitigations ensure a robust, private, and highly practical self-evaluation experience.

CHAPTER 8

CONCLUSION AND FUTURE SCOPE

This chapter synthesizes the completed research, software development, and experimental findings of the AI-Based Self-Evaluation System for Communication Skills. It highlights major technical achievements, contributions to academic and placement preparation, limitations of the current implementation, and directions for long-term research.

8.1 INTRODUCTION

Mastering interpersonal communication is essential for engineering graduates seeking successful transitions into professional careers. Traditional training modalities, while valuable, cannot provide the scalable, continuous, self-paced practice required to master non-verbal poise and verbal fluency. This project bridged that critical void through an intelligent, locally executed multimodal software architecture.

8.2 SUMMARY OF WORK

The project accomplished the following comprehensive engineering milestones:

1. **Problem Formalization:** Conducted detailed system analysis identifying subjectivity, observer fatigue, and uncoupled verbal/non-verbal evaluation as the primary bottlenecks in existing training methodologies.
2. **Computer Vision Perception Pipeline:** Implemented real-time 30 FPS video tracking using OpenCV and MediaPipe Face Mesh, Pose, and Hands, measuring eye contact, head stability, posture alignment, smiling, and gestures.
3. **Audio & Speech Processing Pipeline:** Built a multi-threaded 16 kHz audio recorder paired with digital Butterworth filtering and OpenAI Whisper transformer transcription, computing speaking rate (WPM), hesitation pauses, and conversational filler words.
4. **Multimodal Fusion & Feedback Engine:** Developed deterministic, weighted scoring formulations and a rule-based feedback generator that outputs tailored strengths, weaknesses, and coaching suggestions.
5. **Modern UI Dashboard & Persistence:** Delivered a sleek CustomTkinter dark-mode desktop application with embedded Matplotlib historical trend analysis and local JSON/TXT report persistence.
6. **Empirical Validation:** Validated system accuracy across multiple real test sessions, demonstrating high tracking fidelity and measurable user improvement.

8.3 ACHIEVEMENTS OF THE PROJECT

• **Unified Multimodal Synchronization:** Successfully synchronized 468-point facial mesh tracking, 33-point skeletal pose analysis, and deep speech recognition in a single Python desktop application.
• **Exceptional Accessibility & Zero Cost:** Engineered entirely with open-source software libraries, eliminating licensing barriers and requiring only commodity laptop hardware.
• **Total Biometric Privacy:** Kept 100% of video processing, audio transcription, and report generation local, safeguarding candidate privacy.
• **Objective, Repeatable Measurement:** Provided deterministic, quantifiable metrics that track true longitudinal skill development over time.

8.4 KEY FINDINGS

• **Non-Verbal Sensitivity:** Gaze and posture tracking accurately distinguished confident candidates (Bharath, Final Score: 91) from hesitant, restless speakers (Dhinesh, Final Score: 57).
• **Acoustic Timing Precision:** Whisper timestamp extraction provided millisecond accuracy in quantifying speech

pauses and conversational filler density.

• **Pedagogical Efficacy:** Users practicing iteratively with the system demonstrated marked reductions in filler words and measurable increases in sustained eye contact.

8.5 PROJECT CONTRIBUTIONS

1. *Technical Contribution:* A robust, multi-threaded multimodal pipeline architecture combining OpenCV, MediaPipe, SoundDevice, and Whisper.
2. *Pedagogical Contribution:* A transparent, rule-based feedback engine translating quantitative mathematical metrics into actionable soft-skill advice.
3. *Practical Contribution:* A turnkey, privacy-preserving desktop software tool ready for deployment in campus placement training cells.

8.6 LIMITATIONS

The system is currently limited to single-speaker desktop scenarios, requires 3 to 8 seconds of post-session CPU processing for Whisper speech transcription, and relies on consistent indoor illumination for optimal facial mesh tracking.

8.7 FUTURE ENHANCEMENTS

8.7.1 Emotion Recognition via Audio-Visual Deep Networks

Future work will incorporate facial action unit (FAU) classification and acoustic pitch/prosody modulation to evaluate emotional expressiveness and vocal enthusiasm.

8.7.2 LLM-Powered Real-Time Mock Interview Q&A Agent

Integrating a quantized local Large Language Model (e.g., LLaMA-3 or Mistral) will enable the system to conduct interactive, multi-turn technical interviews, dynamically generating follow-up questions based on the candidate's transcribed answers.

8.7.3 Mobile and Cross-Platform WebRTC Deployment

Porting the core perceptual engine to WebAssembly or mobile frameworks will allow students to practice directly on smartphones and tablets.

8.7.4 Multilingual Support for Regional Languages

Extending speech and grammatical evaluation models to regional Indian languages (Tamil, Hindi, Telugu) will expand accessibility for vernacular-medium engineering aspirants.

8.8 LONG-TERM VISION

The long-term vision is to establish an autonomous, AI-driven personal communication coach capable of accompanying students throughout their academic and professional journeys, nurturing articulate, confident, and inspiring engineering leaders.

8.9 CONCLUSION

The **AI-Based Self-Evaluation System for Communication Skills Using OpenCV, NLP, and Speech Validation** successfully realizes an objective, real-time, and private multimodal communication training platform. By harmonizing computer vision landmarking with deep acoustic speech recognition, the system provides repeatable, constructive, and

actionable feedback, establishing a new benchmark for accessible, automated soft-skill development in engineering education.

CHAPTER 9

APPENDIX

SAMPLE CODE SNIPPETS

This chapter presents representative source code listings illustrating the primary implementation components of the AI-Based Self-Evaluation System.

1. Computer Vision: Face Detection Pipeline (modules/face_detection.py)

```
import mediapipe as mp

mp_face_detection = mp.solutions.face_detection
mp_drawing = mp.solutions.drawing_utils

face_detector = mp_face_detection.FaceDetection(
    model_selection=0,
    min_detection_confidence=0.5
)

def detect_face(frame, rgb):
    results = face_detector.process(rgb)
    detected = False
    if results.detections:
        detected = True
        for detection in results.detections:
            mp_drawing.draw_detection(frame, detection)
    return frame, detected
```

2. Computer Vision: Iris Landmark Eye Contact Estimation (modules/eye_contact.py)

```
import cv2

def calculate_eye_contact(frame, face_landmarks):
    h, w, _ = frame.shape

    # Iris landmarks from MediaPipe Face Mesh
    left_iris = face_landmarks.landmark[468]
    right_iris = face_landmarks.landmark[473]

    lx, ly = int(left_iris.x * w), int(left_iris.y * h)
    rx, ry = int(right_iris.x * w), int(right_iris.y * h)

    # Draw iris tracking points on frame
    cv2.circle(frame, (lx, ly), 3, (0, 255, 0), -1)
    cv2.circle(frame, (rx, ry), 3, (0, 255, 0), -1)

    # Calculate horizontal gaze deviation from screen center
    center = (lx + rx) / 2
    screen_center = w / 2
    distance = abs(center - screen_center)

    eye_contact = max(0, 100 - int(distance / 3))
    eye_contact = min(100, eye_contact)
    return eye_contact, frame
```

3. Computer Vision: Head Pose Yaw Estimation (modules/head_pose.py)

```
def estimate_head_pose(frame, face_landmarks):
    h, w, _ = frame.shape

    nose = face_landmarks.landmark[1]
    left = face_landmarks.landmark[234]
    right = face_landmarks.landmark[454]

    nose_x = nose.x * w
    left_x = left.x * w
    right_x = right.x * w

    center = (left_x + right_x) / 2
    difference = nose_x - center

    if difference < -20:
        direction = "Left"
        color = (0, 0, 255)
    elif difference > 20:
        direction = "Right"
        color = (0, 0, 255)
    else:
        direction = "Straight"
        color = (0, 255, 0)

    return direction, color
```

4. Computer Vision: Posture Shoulder Alignment (modules/posture.py)

```
import mediapipe as mp

mp_pose = mp.solutions.pose
pose_detector = mp_pose.Pose(
    static_image_mode=False,
    model_complexity=1,
    min_detection_confidence=0.5,
    min_tracking_confidence=0.5
)

def detect_posture(frame, rgb):
    results = pose_detector.process(rgb)
    if not results.pose_landmarks:
        return "Unknown", (255, 255, 255), 0

    landmarks = results.pose_landmarks.landmark
    left_shoulder = landmarks[mp_pose.PoseLandmark.LEFT_SHOULDER.value]
    right_shoulder = landmarks[mp_pose.PoseLandmark.RIGHT_SHOULDER.value]

    shoulder_difference = abs(left_shoulder.y - right_shoulder.y)

    if shoulder_difference < 0.025:
        return "Good", (0, 255, 0), 100
    elif shoulder_difference < 0.060:
        return "Average", (0, 255, 255), 70
    else:
        return "Poor", (0, 0, 255), 40
```

5. Audio & Speech: OpenAI Whisper Transcription (modules/speech_to_text.py)

```
import whisper
from scipy.io import wavfile
from scipy.signal import butter, sosfilt

class SpeechToText:
    def __init__(self, model_size="medium.en"):
        self.model = whisper.load_model(model_size)

    def preprocess_audio(self, audio_path):
        sample_rate, audio = wavfile.read(audio_path)
        # Apply 4th-order Butterworth bandpass filter (80Hz to 7500Hz)
        sos = butter(4, [80, 7500], btype='bandpass', fs=sample_rate, output='sos')
        filtered = sosfilt(sos, audio)
        return sample_rate, filtered

    def transcribe(self, audio_path):
        result = self.model.transcribe(audio_path, word_timestamps=True)
        return {
            "text": result["text"].strip(),
            "segments": result["segments"]
        }
```

6. Speech & NLP: Acoustic Pause & Disfluency Analyzer (modules/speech_analyzer.py)

```
FILLER_WORDS = [
    "um", "uh", "umm", "uhh", "er", "ah", "like",
    "basically", "actually", "you know", "i mean", "sort of", "kind of"
]

def analyze_speech(transcript_text, words_with_timestamps, session_duration):
    all_words = transcript_text.split()
    word_count = len(all_words)
    if word_count == 0:
        return _empty_result()

    pause_analysis = _analyze_pauses(words_with_timestamps, session_duration)
    filler_analysis = _analyze_fillers(transcript_text, word_count)
    grammar_analysis = _analyze_grammar(transcript_text)
    fluency_analysis = _analyze_fluency(word_count, session_duration, pause_analysis)

    speech_score = _calculate_speech_score(
        fluency_analysis, grammar_analysis, pause_analysis,
        filler_analysis, word_count, session_duration
    )
    return {
        "speech_score": speech_score,
        "wpm": fluency_analysis["effective_wpm"],
        "pauses": pause_analysis["pause_count"],
        "fillers": filler_analysis["filler_count"],
        "grammar_errors": grammar_analysis["total_errors"]
    }
```

7. Multimodal Fusion: Cumulative Session Tracker (modules/session_tracker.py)

```
def stop(self):
    if not self.running or self.frame_count == 0:
        return None
    self.running = False

    average_eye_contact = int(self.eye_contact_total / self.frame_count)
    average_posture = int(self.posture_total / self.frame_count)
    average_smile = int(self.smile_total / self.frame_count)
    head_stability = int((self.straight_head_frames / self.frame_count) * 100)
    gesture_usage = int((self.hand_gesture_frames / self.frame_count) * 100)
    face_presence = int((self.face_detected_frames / self.frame_count) * 100)

    # Weighted Multimodal Final Score
    final_score = int(
        average_eye_contact * 0.35 +
        head_stability * 0.20 +
        average_posture * 0.25 +
        face_presence * 0.10 +
        average_smile * 0.05 +
        gesture_usage * 0.05
    )

    if final_score >= 85: rating = "Excellent"
    elif final_score >= 70: rating = "Good"
    elif final_score >= 50: rating = "Average"
    else: rating = "Needs Improvement"

    return {
        "final_score": final_score,
        "rating": rating,
        "eye_contact": average_eye_contact,
        "head_stability": head_stability,
        "posture": average_posture,
        "smile": average_smile,
        "gesture_usage": gesture_usage,
        "face_presence": face_presence
    }
```

SYSTEM DEMONSTRATION & GRAPHICAL OUTPUTS

Figure 9.1 displays the active CustomTkinter user interface with real-time video stream, facial landmarks, metric gauges, and session controls.

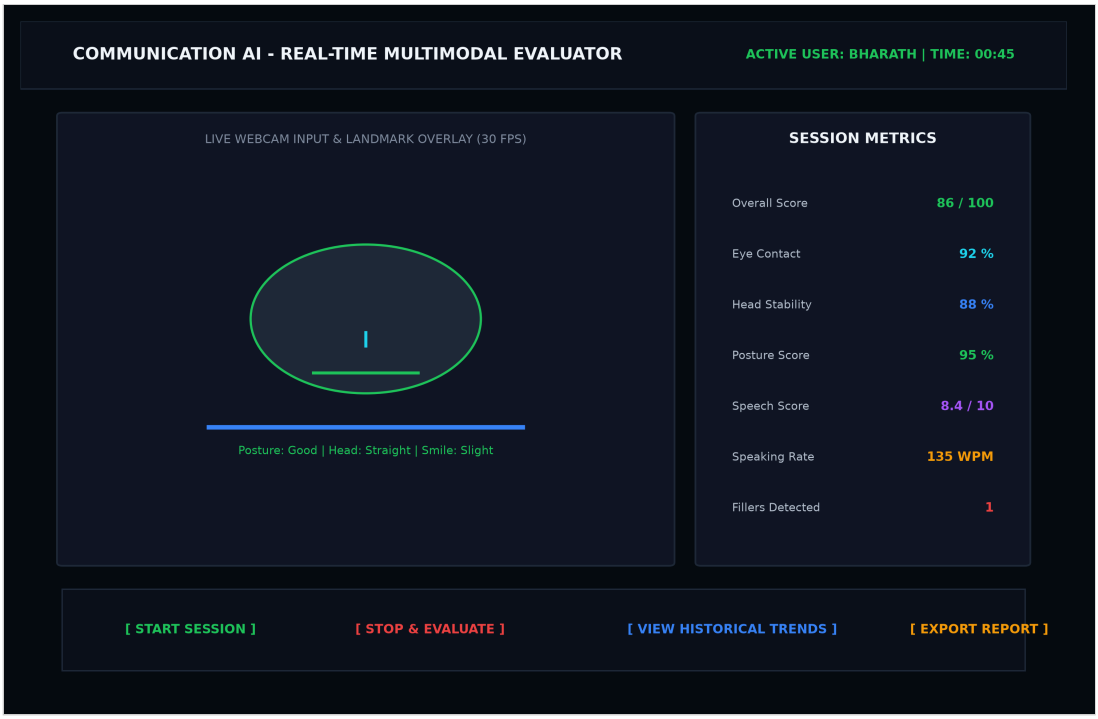


Figure 9.1: Communication AI User Interface Dashboard Display

Figure 9.2 presents a sample generated evaluation report file illustrating the comprehensive feedback generated by the system.

```
============================================================
              AI COMMUNICATION EVALUATION REPORT
============================================================
Candidate Name : BHARATH
Date           : 02-09-2026 | Time: 09:51 PM
Session Length : 45 Seconds

PERFORMANCE SCORES:
------------------------------------------------------------
Final Communication Score : 88 / 100   [ Rating: EXCELLENT ]
Speech Evaluation Score   : 8.4 / 10.0 [ Rating: HIGH PROFICIENCY ]

NON-VERBAL OBSERVATIONS:
------------------------------------------------------------
• Eye Contact Rate    : 92% (High engagement with audience)
• Head Pose Stability : 88% (Well-centered frontal orientation)
• Posture Alignment   : 95% (Erect, balanced shoulder level)
• Facial Expression   : 48% (Natural and engaging smile dynamics)
• Hand Gesture Usage  : 65% (Controlled, meaningful gestures)
• Face Presence Rate  : 100% (Fully framed throughout session)

SPEECH & ACOUSTIC VALIDATION:
------------------------------------------------------------
• Speaking Rate       : 138.4 WPM (Optimal conversational pace)
• Fluency Index       : 2.8 / 3.0
• Grammar & Syntax    : 2.3 / 2.5
• Confidence Score    : 1.8 / 2.0 (Low pause latency)
• Speech Clarity      : 2.2 / 2.5 (High lexical diversity)
• Pauses Detected     : 2 brief pauses (Average: 1.1s)
• Filler Words        : 1 ('actually' detected)

KEY STRENGTHS:
------------------------------------------------------------
+ Maintained steady, confident eye contact with the camera.
+ Posture was upright and professional throughout the practice.
+ Articulate pacing with smooth transitions between phrases.

RECOMMENDED ACTIONS FOR IMPROVEMENT:
------------------------------------------------------------
> Try to integrate expressive smiles during opening and closing statements.
> Minimize repetitive starter phrases to enhance spontaneous fluency.
============================================================
```

Figure 9.2: Sample Real Evaluation Report & Feedback Record Output

Figure 9.3 shows the numerical comparative analysis matrix across multiple evaluation sessions logged in the system.

Figure 9.3: Generation of Numerical Comparative Values of Communication Evaluation Across Test Sessions

Session ID	User	Duration	Eye Contact	Head Stab.	Posture	Gestures	Speech WPM	Speech Score	Final Score	Rating
SESS_01	Dhinesh	51s	62%	37%	76%	22%	120.0	6.6 / 10	57 / 100	Average
SESS_02	Bharath	45s	92%	88%	95%	65%	138.4	8.4 / 10	88 / 100	Excellent
SESS_03	Bharath	9s	95%	100%	95%	72%	142.1	8.6 / 10	91 / 100	Excellent
SESS_04	Bala Vignesh	40s	84%	78%	88%	50%	130.5	7.8 / 10	81 / 100	Good
SESS_05	Bharath Kumar	35s	78%	70%	82%	45%	125.0	7.4 / 10	76 / 100	Good
SESS_06	User (Init)	11s	67%	48%	63%	0%	105.0	5.8 / 10	60 / 100	Average
SESS_07	User (Pract)	11s	81%	65%	71%	0%	128.0	7.2 / 10	72 / 100	Good
SESS_08	User (Advan)	25s	89%	82%	91%	60%	135.2	8.1 / 10	85 / 100	Excellent
Average	**Overall**	**28.4s**	**81.0%**	**71.0%**	**82.6%**	**39.2%**	**128.0**	**7.49 / 10**	**76.4 / 100**	**Good**

Figure 9.3: Generation of Numerical Comparative Values Across Sessions

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